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Respiratory pressure therapy device

Abstract

Apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder includes a first chamber, a second chamber, at least one inlet tube structured and configured to allow ambient air to enter the first chamber, at least one flow tube structured and configured to allow air to pass from the first chamber to the second chamber, and a blower structured and configured to produce a flow of air at positive pressure. The blower is positioned in the first chamber and structured and configured to receive air from the second chamber. The blower includes a housing structured and configured to sealingly separate air flow through an interior of the housing from the first chamber. The at least one inlet tube is axially spaced from the at least one flow tube.

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Background/Summary

1 CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/027,831, filed Sep. 22, 2020, which is a continuation of U.S. application Ser. No. 15/741,572, filed Jan. 3, 2018, now U.S. Pat. No. 10,814,083, which is the U.S. national phase of International Application No. PCT/IB2016/051434, filed Mar. 14, 2016, which designated the U.S. and claims the benefit of U.S. Provisional Patent Application No. 62/189,483, filed Jul. 7, 2015, the entire contents of each of which are hereby incorporated by reference.

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2 BACKGROUND OF THE TECHNOLOGY

2.1 Field of the Technology

(2) The present technology relates to one or more of the detection, diagnosis, treatment, prevention and amelioration of respiratory-related disorders. The present technology also relates to medical devices or apparatus, and their use.

2.2 Description of the Related Art

(3) 2.2.1 Human Respiratory System and its Disorders

(4) The respiratory system of the body facilitates gas exchange. The nose and mouth form the entrance to the airways of a patient.

(5) The airways include a series of branching tubes, which become narrower, shorter and more numerous as they penetrate deeper into the lung. The prime function of the lung is gas exchange, allowing oxygen to move from the air into the venous blood and carbon dioxide to move out. The trachea divides into right and left main bronchi, which further divide eventually into terminal bronchioles. The bronchi make up the conducting airways, and do not take part in gas exchange. Further divisions of the airways lead to the respiratory bronchioles, and eventually to the alveoli. The alveolated region of the lung is where the gas exchange takes place, and is referred to as the respiratory zone. See "*Respiratory Physiology*", by John B. West, Lippincott Williams & Wilkins, 9th edition published 2011.

(6) A range of respiratory disorders exist. Certain disorders may be characterised by particular events, e.g. apneas, hypopneas, and hyperpneas.

(7) A range of therapies have been used to treat or ameliorate such conditions. Furthermore, otherwise healthy individuals may take advantage of such therapies to prevent respiratory disorders from arising. However, these have a number of shortcomings.

(8) 2.2.2 Therapy

(9) Continuous Positive Airway Pressure (CPAP) therapy has been used to treat Obstructive Sleep Apnea (OSA). The mechanism of action is that continuous positive airway pressure acts as a pneumatic splint and may prevent upper airway occlusion, such as by pushing the soft palate and tongue forward and away from the posterior oropharyngeal wall. Treatment of OSA by CPAP therapy may be voluntary, and hence patients may elect not to comply with therapy if they find devices used to provide such therapy one or more of: uncomfortable, difficult to use, expensive and aesthetically unappealing.

(10) Non-invasive ventilation (NIV) provides ventilatory support to a patient through the upper airways to assist the patient breathing and/or maintain adequate oxygen levels in the body by doing some or all of the work of breathing. The ventilatory support is provided via a non-invasive patient interface. NIV has been used to treat CSR and respiratory insufficiency, in forms such as OHS, COPD, MD and Chest Wall disorders. In some forms, the comfort and effectiveness of these therapies may be improved.

(11) Invasive ventilation (IV) provides ventilatory support to patients that are no longer able to effectively breathe themselves and may be provided using a tracheostomy tube. In some forms, the comfort and effectiveness of these therapies may be improved.

(12) 2.2.3 Treatment Systems

(13) These therapies may be provided by a treatment system or device. Such systems and devices may also be used to diagnose a condition without treating it.

(14) A treatment system may comprise a Respiratory Pressure Therapy Device (RPT device), an air circuit, a humidifier, a patient interface, and data management.

(15) Another form of treatment system is a mandibular repositioning device.

(16) 2.2.3.1 Patient Interface

(17) A patient interface may be used to interface respiratory equipment to its wearer, for example by providing a flow of air to an entrance to the airways. The flow of air may be provided via a mask to the nose and/or mouth, a tube to the mouth or a tracheostomy tube to the trachea of a patient. Depending upon the therapy to be applied, the patient interface may form a seal, e.g., with a region of the patient's face, to facilitate the delivery of gas at a pressure at sufficient variance with ambient pressure to effect therapy, e.g., at a positive pressure of about 10 cmH₂O relative to ambient pressure. For other forms of therapy, such as the delivery of oxygen, the patient interface may not include a seal sufficient to facilitate delivery to the airways of a supply of gas at a positive

pressure of about 10 cmH.sub.2O.

(18) Certain other mask systems may be functionally unsuitable for the present field. For example, purely ornamental masks may be unable to maintain a suitable pressure. Mask systems used underwater swimming or diving may be configured to guard against ingress of water from an external higher pressure, but not to maintain air internally at a higher pressure than ambient.

(19) Certain masks may be impractical for use while sleeping, e.g. for sleeping while lying on one's side in bed with a head on a pillow.

(20) CPAP therapy is highly effective to treat certain respiratory disorders, provided patients comply with therapy. If a mask is uncomfortable, or difficult to use a patient may not comply with therapy. Since it is often recommended that a patient regularly wash their mask, if a mask is difficult to clean (e.g., difficult to assemble or disassemble), patients may not clean their mask and this may impact on patient compliance.

(21) For these reasons, patient interfaces for delivery of CPAP during sleep form a distinct field.

(22) 2.2.3.1.1 Seal-Forming Portion

(23) Patient interfaces may include a seal-forming portion. Since it is in direct contact with the patient's face, the shape and configuration of the seal-forming portion can have a direct impact the effectiveness and comfort of the patient interface.

(24) A patient interface may be partly characterised according to the design intent of where the seal-forming portion is to engage with the face in use. In one form of patient interface, a seal-forming portion may comprise two sub-portions to engage with respective left and right nares. In one form of patient interface, a seal-forming portion may comprise a single element that surrounds both nares in use. Such single element may be designed to for example overlay an upper lip region and a nasal bridge region of a face. In one form of patient interface a seal-forming portion may comprise an element that surrounds a mouth region in use, e.g. by forming a seal on a lower lip region of a face. In one form of patient interface, a seal-forming portion may comprise a single element that surrounds both nares and a mouth region in use. These different types of patient interfaces may be known by a variety of names by their manufacturer including nasal masks, full-face masks, nasal pillows, nasal puffs and oro-nasal masks.

(25) 2.2.3.1.2 Positioning and Stabilising

(26) A seal-forming portion of a patient interface used for positive air pressure therapy is subject to the corresponding force of the air pressure to disrupt a seal. Thus a variety of techniques have been used to position the seal-forming portion, and to maintain it in sealing relation with the appropriate portion of the face.

(27) 2.2.3.2 Vent Technologies

(28) Some forms of respiratory treatment systems may include a vent to allow the washout of exhaled carbon dioxide. The vent may allow a flow of gas from an interior space of the patient interface, e.g., the plenum chamber, to an exterior of the patient interface, e.g., to ambient. The vent may comprise an orifice and gas may flow through the orifice in use of the mask. Many such vents are noisy. Others may become blocked in use and thus provide insufficient washout. Some vents may be disruptive of the sleep of a bed partner **1100** of the patient **1000**, e.g. through noise or focussed airflow.

(29) ResMed Limited has developed a number of improved mask vent technologies. See International Patent Application Publication No. WO 1998/034,665; International Patent Application Publication No. WO 2000/078,381; U.S. Pat. No. 6,581,594; US Patent Application Publication No. US 2009/0050156; US Patent Application Publication No. 2009/0044808.

(30) TABLE-US-00001 Table of noise of prior masks (ISO 17510-2: 2007, 10 cmH.sub.2O pressure at 1 m) A-weighted sound A-weighted sound power level dB(A) pressure dB(A) Year Mask name Mask type (uncertainty) (uncertainty) (approx.) Glue-on (*) nasal 50.9 42.9 1981 ResCare standard (*) nasal 31.5 23.5 1993 ResMed Mirage TM (*) nasal 29.5 21.5 1998 ResMed UltraMirage TM nasal 36 (3) 28 (3) 2000 ResMed Mirage Activa TM nasal 32 (3) 24 (3) 2002

ResMed Mirage Micro™ nasal 30 (3) 22 (3) 2008 ResMed Mirage™ SoftGel nasal 29 (3) 22 (3) 2008 ResMed Mirage™ FX nasal 26 (3) 18 (3) 2010 ResMed Mirage Swift™ (*) nasal pillows 37 29 2004 ResMed Mirage Swift™ II nasal pillows 28 (3) 20 (3) 2005 ResMed Mirage Swift™ LT nasal pillows 25 (3) 17 (3) 2008 ResMed AirFit P10 nasal pillows 21 (3) 13 (3) 2014 ((*) one specimen only, measured using test method specified in ISO 3744 in CPAP mode at 10 cmH.sub.2O) Sound pressure values of a variety of objects are listed below

(31) TABLE-US-00002 A-weighted sound Object pressure dB(A) Notes Vacuum cleaner: Nilfisk Walter 68 ISO 3744 at Broadly Litter Hog: B+ Grade 1 m distance Conversational speech 60 1 m distance Average home 50 Quiet library 40 Quiet bedroom at night 30 Background in TV studio 20
2.2.3.3 Respiratory Pressure Therapy (RPT) Device

(32) Air pressure generators are known in a range of applications, e.g. industrial-scale ventilation systems. However, air pressure generators for medical applications have particular requirements not fulfilled by more generalised air pressure generators, such as the reliability, size and weight requirements of medical devices. In addition, even devices designed for medical treatment may suffer from shortcomings, pertaining to one or more of: comfort, noise, ease of use, efficacy, size, weight, manufacturability, cost, and reliability.

(33) An example of the special requirements of certain RPT devices is acoustic noise.

(34) Table of noise output levels of prior RPT devices (one specimen only, measured using test method specified in ISO 3744 in CPAP mode at 10 cmH.sub.2O).

(35) TABLE-US-00003 A-weighted sound Year RPT Device name power level dB(A) (approx.) C-Series Tango™ 31.9 2007 C-Series Tango™ with Humidifier 33.1 2007 S8 Escape™ II 30.5 2005 S8 Escape™ II with H4i™ Humidifier 31.1 2005 S9 AutoSet™ 26.5 2010 S9 AutoSet™ with H5i Humidifier 28.6 2010

(36) One known RPT device used for treating sleep disordered breathing is the S9 Sleep Therapy System, manufactured by ResMed Limited. Another example of an RPT device is a ventilator. Ventilators such as the ResMed Stellar™ Series of Adult and Paediatric Ventilators may provide support for invasive and non-invasive non-dependent ventilation for a range of patients for treating a number of conditions such as but not limited to NMD, OHS and COPD.

(37) The ResMed Elisée™ 150 ventilator and ResMed VS III™ ventilator may provide support for invasive and non-invasive dependent ventilation suitable for adult or paediatric patients for treating a number of conditions. These ventilators provide volumetric and barometric ventilation modes with a single or double limb circuit. RPT devices typically comprise a pressure generator, such as a motor-driven blower or a compressed gas reservoir, and are configured to supply a flow of air to the airway of a patient. In some cases, the flow of air may be supplied to the airway of the patient at positive pressure. The outlet of the RPT device is connected via an air circuit to a patient interface such as those described above.

(38) The designer of a device may be presented with an infinite number of choices to make. Design criteria often conflict, meaning that certain design choices are far from routine or inevitable. Furthermore, the comfort and efficacy of certain aspects may be highly sensitive to small, subtle changes in one or more parameters.

(39) 2.2.3.4 Humidifier

(40) Delivery of a flow of air without humidification may cause drying of airways. The use of a humidifier with an RPT device and the patient interface produces humidified gas that minimizes drying of the nasal mucosa and increases patient airway comfort. In addition in cooler climates, warm air applied generally to the face area in and about the patient interface is more comfortable than cold air. A range of artificial humidification devices and systems are known, however they may not fulfil the specialised requirements of a medical humidifier.

(41) Medical humidifiers are used to increase humidity and/or temperature of the flow of air in relation to ambient air when required, typically where the patient may be asleep or resting (e.g. at a hospital). A medical humidifier for bedside placement may be small. A medical humidifier may be

configured to only humidify and/or heat the flow of air delivered to the patient without humidifying and/or heating the patient's surroundings. Room-based systems (e.g. a sauna, an air conditioner, or an evaporative cooler), for example, may also humidify air that is breathed in by the patient, however those systems would also humidify and/or heat the entire room, which may cause discomfort to the occupants. Furthermore medical humidifiers may have more stringent safety constraints than industrial humidifiers

(42) While a number of medical humidifiers are known, they can suffer from one or more shortcomings. Some medical humidifiers may provide inadequate humidification, some are difficult or inconvenient to use by patients.

(43) 2.2.3.5 Data Management

(44) There may be clinical reasons to obtain data to determine whether the patient prescribed with respiratory therapy has been “compliant”, e.g. that the patient has used their RPT device according to certain a “compliance rule”. One example of a compliance rule for CPAP therapy is that a patient, in order to be deemed compliant, is required to use the RPT device for at least four hours a night for at least 21 of 30 consecutive days. In order to determine a patient's compliance, a provider of the RPT device, such as a health care provider, may manually obtain data describing the patient's therapy using the RPT device, calculate the usage over a predetermined time period, and compare with the compliance rule. Once the health care provider has determined that the patient has used their RPT device according to the compliance rule, the health care provider may notify a third party that the patient is compliant.

(45) There may be other aspects of a patient's therapy that would benefit from communication of therapy data to a third party or external system.

(46) Existing processes to communicate and manage such data can be one or more of costly, time-consuming, and error-prone.

3 BRIEF SUMMARY OF THE TECHNOLOGY

(47) The present technology is directed towards providing medical devices used in the diagnosis, amelioration, treatment, or prevention of respiratory disorders having one or more of improved comfort, cost, efficacy, ease of use and manufacturability.

(48) A first aspect of the present technology relates to apparatus used in the diagnosis, amelioration, treatment or prevention of a respiratory disorder.

(49) Another aspect of the present technology relates to methods used in the diagnosis, amelioration, treatment or prevention of a respiratory disorder.

(50) An aspect of certain forms of the present technology is to provide methods and/or apparatus that improve the compliance of patients with respiratory therapy.

(51) An aspect of the present technology relates to an RPT device structured and configured to reduce noise output, e.g., while maintaining a relatively small size.

(52) An aspect of the present technology relates to an RPT device including an inlet having a plurality of inlet tubes, e.g., arranged in an array, to reduce noise output.

(53) An aspect of the present technology relates to an RPT device including a first chamber, a second chamber, an inlet tube array to deliver air from an inlet to the first chamber, and a flow tube array to deliver air from the first chamber to the second chamber. In an example, the RPT device includes a blower positioned in the first chamber and structured and configured to receive air from the second chamber. In an example, the RPT device includes a flow rate sensor structured and configured to measure a first pressure in the first chamber and a second pressure in the second chamber to determine an air flow rate. In an example, the inlet tube array is axially spaced from and generally parallel to the flow tube array, however one or more tubes of the inlet tube array are not co-axial with one or more tubes of the flow tube array.

(54) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including a first chamber, a second chamber, at least one inlet tube structured and configured to allow ambient air to

enter the first chamber, at least one flow tube structured and configured to allow air to pass from the first chamber to the second chamber, and a blower structured and configured to produce a flow of air at positive pressure. The blower is positioned in the first chamber and structured and configured to receive air from the second chamber. The blower includes a housing structured and configured to sealingly separate air flow through an interior of the housing from the first chamber. The at least one inlet tube is axially spaced from the at least one flow tube.

(55) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including at least one chamber and an inlet tube array including a plurality of inlet tubes structured and configured to allow ambient air to enter the at least one chamber. At least one of the plurality of inlet tubes is arranged adjacent to another inlet tube such that the adjacent inlet tubes include at least one common side wall.

(56) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including at least one chamber, an inlet tube array including a plurality of inlet tubes structured and configured to allow ambient air to enter the at least one chamber, and a flow tube array including a plurality of flow tubes structured and configured to allow air to exit the at least one chamber. The inlet tube array is axially spaced apart from the flow tube array. The inlet tubes of the inlet tube array include axes that are arranged substantially parallel to axes of the flow tubes of the flow tube array. At least one of the inlet tubes includes an axis that is axially offset from an axis of at least one of the flow tubes.

(57) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including at least one chamber, a blower structured and configured to produce a flow of air at positive pressure, wherein the blower is located downstream of the at least one chamber, and a plate assembly including a base plate that defines a wall of the at least one chamber, at least one tube structured and configured to allow air to enter the at least one chamber, and a blower suspension to support the blower.

(58) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including a blower, a housing including an upper portion and a lower portion and defining at least one chamber, a first plate assembly including a support for the blower and at least one tube structured and configured to allow air to enter the at least one chamber, and a second plate assembly including a support for the blower and at least one tube structured and configured to allow air to exit the at least one chamber. The upper and lower portions are engageable or separable in a generally normal direction with respect to an axis of the at least one tube of the first plate assembly.

(59) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including at least one chamber, a blower structured and configured to produce a flow of air at positive pressure, a first plate assembly including a base plate that defines a wall of the at least one chamber and an outlet end suspension to support the blower adjacent a blower outlet of the blower, and a second plate assembly including a base plate that defines a wall of the at least one chamber and an inlet end suspension to support the blower adjacent a blower inlet of the blower.

(60) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including a chamber including an air flow path, a blower positioned in the chamber, a printed circuit board positioned exterior the chamber, and a connector structured and configured to electrically connect the blower and the printed circuit board. The connector is arranged to pass through the chamber while not being positioned directly in the air flow path.

(61) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including: a housing

including an upper portion, a lower portion, and an intermediate portion between upper and lower portions, the lower portion and the intermediate portion defining a first chamber portion including an air flow path, and the upper portion and the intermediate portion defining a second chamber exterior the air flow path; a blower positioned in the first chamber; and a printed circuit board positioned in the second chamber.

(62) Another aspect of the present technology relates to a plate assembly for an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including a base plate structured and configured to define a wall of a chamber, a plurality of tubes provided to the base plate and structured and configured to allow air to enter or exit the chamber, and a blower suspension provided to the base plate and structured and configured to support an end of a blower.

(63) Another aspect of the present technology relates to an apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder including a housing including a housing inlet and a housing outlet and internal components provided to the housing. The internal components include at least a blower structured and configured to produce a flow of air at positive pressure. The housing and the internal components cooperate to form an air flow path having a general U-shape extending from the housing inlet to the housing outlet. In an example, the blower includes a blower inlet and a blower outlet, and the air flow path extends from the housing inlet to the blower inlet and from blower outlet to the housing outlet; the housing includes an end portion that provides both the housing inlet and the housing outlet; the blower is provided along a leg of the U-shape, e.g., along an outlet leg of the U-shape extending from the housing outlet; the blower includes an axis that is generally co-linear with an axis of the outlet leg of the U-shape; and/or the U-shaped air flow path extends substantially in the same plane.

(64) An aspect of certain forms of the present technology is a medical device that is easy to use, e.g. by a person who does not have medical training, by a person who has limited dexterity or by a person with limited experience in using this type of medical device.

(65) An aspect of one form of the present technology is a portable RPT device that may be carried by a person, e.g., around the home of the person.

(66) Of course, portions of the aspects may form sub-aspects of the present technology. Also, various ones of the sub-aspects and/or aspects may be combined in various manners and also constitute additional aspects or sub-aspects of the present technology.

(67) Other features of the technology will be apparent from consideration of the information contained in the following detailed description, abstract, drawings and claims.

Description

4 BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present technology is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings, in which like reference numerals refer to similar elements including:

4.1 Treatment Systems

(2) FIG. 1A shows a system including a patient **1000** wearing a patient interface **3000**, in the form of a nasal pillows, receiving a supply of air at positive pressure from an RPT device **4000**. Air from the RPT device is humidified in a humidifier **5000**, and passes along an air circuit **4170** to the patient **1000**. A bed partner **1100** is also shown.

(3) FIG. 1B shows a system including a patient **1000** wearing a patient interface **3000**, in the form of a nasal mask, receiving a supply of air at positive pressure from an RPT device **4000**. Air from the RPT device is humidified in a humidifier **5000**, and passes along an air circuit **4170** to the patient **1000**.

(4) FIG. 1C shows a system including a patient **1000** wearing a patient interface **3000**, in the form of a full-face mask, receiving a supply of air at positive pressure from an RPT device **4000**. Air from the RPT device is humidified in a humidifier **5000**, and passes along an air circuit **4170** to the patient **1000**.

4.2 Respiratory System and Facial Anatomy

(5) FIG. 2 shows an overview of a human respiratory system including the nasal and oral cavities, the larynx, vocal folds, oesophagus, trachea, bronchus, lung, alveolar sacs, heart and diaphragm.

4.3 Patient Interface

(6) FIG. 3 shows a patient interface in the form of a nasal mask in accordance with one form of the present technology.

4.4 RPT Device

(7) FIG. 4A is a perspective view of an RPT device in accordance with one form of the present technology.

(8) FIG. 4B is another perspective view of the RPT device of FIG. 4A.

(9) FIG. 4C is a perspective view of the RPT device of FIG. 4A, with an intermediate cover and portions of the housing removed.

(10) FIG. 4D is another perspective view of the RPT device of FIG. 4A, with an intermediate cover and portions of the housing removed.

(11) FIG. 4E is a top view of the RPT device of FIG. 4A, with an intermediate cover and portions of the housing removed.

(12) FIG. 4F is an exploded view of components of the RPT device shown in FIGS. 4C to 4E.

(13) FIG. 4G is a further exploded view of components of the RPT device shown in FIGS. 4C to 4E.

(14) FIG. 4H is an exploded view of the RPT device shown in FIG. 4A.

(15) FIG. 4I is a cross-sectional view of the RPT device shown in FIG. 4A.

(16) FIG. 4J is an exploded and cross-sectional view of the RPT device shown in FIG. 4A.

(17) FIG. 4K is an enlarged portion of the RPT device of FIG. 4J.

(18) FIG. 4L is an enlarged portion of the RPT device of FIG. 4J.

(19) FIG. 4M is a perspective view of a first plate assembly for an RPT device in accordance with one form of the present technology.

(20) FIG. 4N is a cross-sectional view of the first plate assembly shown in FIG. 4M.

(21) FIG. 4O is a perspective view of a second plate assembly for an RPT device in accordance with one form of the present technology.

(22) FIG. 4P is another perspective view of the second plate assembly shown in FIG. 4O.

(23) FIG. 4Q is a top view showing the arrangement of first and second plate assemblies for an RPT device in accordance with one form of the present technology.

(24) FIG. 4R is a cross-sectional view through line 4R-4R of FIG. 4Q.

(25) FIG. 4S is a schematic view of an inlet tube array for an RPT device in accordance with one form of the present technology.

(26) FIG. 4T is a schematic view of an inlet tube array for an RPT device in accordance with one form of the present technology.

(27) FIG. 4U is a schematic view of an inlet tube array for an RPT device in accordance with one form of the present technology.

(28) FIG. 4V is a schematic diagram of the algorithms implemented in an RPT device in accordance with one form of the present technology.

(29) FIG. 4W1 is a perspective view of an external battery for powering an RPT device in accordance with one form of the present technology.

(30) FIG. 4W2 is another perspective view of the external battery shown in FIG. 4W1.

(31) FIG. 4X1 is a perspective view of the external battery of FIG. 4W1 being engaged with the RPT device of FIG. 4A in accordance with one form of the present technology.

(32) FIG. 4X2 is another perspective view of the external battery and RPT device shown in FIG. 4X1.

(33) FIG. 4Y1 is a perspective view of the external battery of FIG. 4W1 engaged with the RPT device of FIG. 4A in accordance with one form of the present technology.

(34) FIG. 4Y2 is another perspective view of the external battery and RPT device shown in FIG. 4Y1.

4.5 Humidifier

(35) FIG. 5A shows an isometric view of a humidifier in accordance with one form of the present technology.

(36) FIG. 5B shows an isometric view of a humidifier in accordance with one form of the present technology, showing a humidifier reservoir **5110** removed from the humidifier reservoir dock **5130**.

(37) FIG. 5C shows a schematic of a humidifier in accordance with one form of the present technology.

4.6 Breathing Waveforms

(38) FIG. 6 shows a model typical breath waveform of a person while sleeping.

5 DETAILED DESCRIPTION OF EXAMPLES OF THE TECHNOLOGY

(39) Before the present technology is described in further detail, it is to be understood that the technology is not limited to the particular examples described herein, which may vary. It is also to be understood that the terminology used in this disclosure is for the purpose of describing only the particular examples discussed herein, and is not intended to be limiting.

(40) The following description is provided in relation to various examples which may share one or more common characteristics and/or features. It is to be understood that one or more features of any one example may be combinable with one or more features of another example or other examples. In addition, any single feature or combination of features in any of the examples may constitute a further example.

5.1 Therapy

(41) In one form, the present technology comprises a method for treating a respiratory disorder comprising the step of applying positive pressure to the entrance of the airways of a patient **1000**.

(42) In certain examples of the present technology, a supply of air at positive pressure is provided to the nasal passages of the patient via one or both nares.

(43) In certain examples of the present technology, mouth breathing is limited, restricted or prevented.

5.2 Treatment Systems

(44) In one form, the present technology comprises an apparatus or device for treating a respiratory disorder. The apparatus or device may comprise an RPT device **4000** or **8000** for supplying pressurised air to the patient **1000** via an air circuit **4170** to a patient interface **3000**.

5.3 Patient Interface

(45) As shown in FIG. 3, a non-invasive patient interface **3000** in accordance with one aspect of the present technology comprises the following functional aspects: a seal-forming structure **3100**, a plenum chamber **3200**, a positioning and stabilising structure **3300**, a vent **3400**, one form of connection port **3600** for connection to air circuit **4170**, and a forehead support **3700**. In some forms a functional aspect may be provided by one or more physical components. In some forms, one physical component may provide one or more functional aspects. In use the seal-forming structure **3100** is arranged to surround an entrance to the airways of the patient so as to facilitate the supply of air at positive pressure to the airways.

(46) 5.3.1 Seal-Forming Structure

(47) In one form of the present technology, a seal-forming structure **3100** provides a seal-forming surface, and may additionally provide a cushioning function.

(48) A seal-forming structure **3100** in accordance with the present technology may be constructed from a soft, flexible, resilient material such as silicone.

- (49) In one form the seal-forming portion of the non-invasive patient interface **3000** comprises a pair of nasal puffs, or nasal pillows, each nasal puff or nasal pillow being constructed and arranged to form a seal with a respective naris of the nose of a patient.
- (50) In one form, the non-invasive patient interface **3000** comprises a seal-forming portion that forms a seal in use on an upper lip region (that is, the lip superior) of the patient's face.
- (51) In one form the non-invasive patient interface **3000** comprises a seal-forming portion that forms a seal in use on a chin-region of the patient's face.
- (52) 5.3.2 Plenum Chamber
- (53) The plenum chamber **3200** has a perimeter that is shaped to be complementary to the surface contour of the face of an average person in the region where a seal will form in use. In use, a marginal edge of the plenum chamber **3200** is positioned in close proximity to an adjacent surface of the face. Actual contact with the face is provided by the seal-forming structure **3100**. The seal-forming structure **3100** may extend in use about the entire perimeter of the plenum chamber **3200**.
- (54) 5.3.3 Positioning and Stabilising Structure
- (55) The seal-forming structure **3100** of the patient interface **3000** of the present technology may be held in sealing position in use by the positioning and stabilising structure **3300**.
- (56) In one form of the present technology, a positioning and stabilising structure **3300** is provided that is configured in a manner consistent with being worn by a patient while sleeping. In one example the positioning and stabilising structure **3300** has a low profile, or cross-sectional thickness, to reduce the perceived or actual bulk of the apparatus. In one example, the positioning and stabilising structure **3300** comprises at least one strap having a rectangular cross-section. In one example the positioning and stabilising structure **3300** comprises at least one flat strap.
- (57) In one form of the present technology, a positioning and stabilising structure **3300** comprises a strap constructed from a laminate of a fabric patient-contacting layer, a foam inner layer and a fabric outer layer. In one form, the foam is porous to allow moisture, (e.g., sweat), to pass through the strap. In one form, the fabric outer layer comprises loop material to engage with a hook material portion.
- (58) In certain forms of the present technology, a positioning and stabilising structure **3300** comprises a strap that is extensible, e.g. resiliently extensible. For example the strap may be configured in use to be in tension, and to direct a force to draw a cushion into sealing contact with a portion of a patient's face. In an example the strap may be configured as a tie.
- (59) In certain forms of the present technology, a positioning and stabilising structure **3300** comprises a strap that is bendable and e.g. non-rigid. An advantage of this aspect is that the strap is more comfortable for a patient to lie upon while the patient is sleeping.
- (60) 5.3.4 Vent
- (61) In one form, the patient interface **3000** includes a vent **3400** constructed and arranged to allow for the washout of exhaled gases, e.g. carbon dioxide.
- (62) One form of vent **3400** in accordance with the present technology comprises a plurality of holes, for example, about 20 to about 80 holes, or about 40 to about 60 holes, or about 45 to about 55 holes.
- (63) The vent **3400** may be located in the plenum chamber **3200**. Alternatively, the vent **3400** is located in a decoupling structure, e.g., a swivel.
- (64) 5.3.5 Decoupling Structure(s)
- (65) In one form the patient interface **3000** includes at least one decoupling structure, for example, a swivel or a ball and socket.
- (66) 5.3.6 Connection Port
- (67) Connection port **3600** allows for connection to the air circuit **4170**.
- (68) 5.3.7 Forehead Support
- (69) In one form, the patient interface **3000** includes a forehead support **3700**.
- (70) 5.3.8 Anti-Asphyxia Valve

(71) In one form, the patient interface **3000** includes an anti-asphyxia valve.

(72) 5.3.9 Ports

(73) In one form of the present technology, a patient interface **3000** includes one or more ports that allow access to the volume within the plenum chamber **3200**. In one form this allows a clinician to supply supplemental oxygen. In one form, this allows for the direct measurement of a property of gases within the plenum chamber **3200**, such as the pressure.

5.4 RPT Device

(74) An RPT device in accordance with one aspect of the present technology comprises mechanical and pneumatic components, electrical components and is configured to execute one or more algorithms.

(75) FIGS. **4A** to **4R** illustrate an RPT device **8000** according to an example of the present technology. As illustrated, the RPT device **8000** includes an external housing **8010** including an upper or top housing portion **8012**, a lower or bottom housing portion **8014**, a first end portion **8016** including a housing inlet **8018** and a housing outlet **8020**, and a second end portion **8022** providing an end cover including faceplate **8024**.

(76) The housing **8010** supports and/or surrounds internal components of the RPT device **8000** including blower **8030**, intermediate cover **8040**, first plate assembly **8050**, second plate assembly **8060**, and a printed circuit board assembly (PCBA) **8070** including a main printed circuit board (main PCB) **8072** and a secondary printed circuit board (secondary PCB) **8074**.

(77) The housing **8010** and internal components of the RPT device **8000** cooperate to form the pneumatic air flow path or pneumatic block that extends from the housing inlet **8018** to the blower inlet **8032** of the blower **8030** and from the blower outlet **8034** of the blower **8030** to the housing outlet **8020**.

(78) In an example, the housing and the internal components cooperate to form an air flow path having a general U-shape extending from the housing inlet to the housing outlet. For example, the U-shaped air flow path may include an inlet leg extending from the housing inlet, an outlet leg extending from the housing outlet, and a connecting leg that interconnects the inlet leg and the outlet leg. In an example, the inlet leg and the outlet leg are generally parallel to one another. In an example, the blower is provided along a leg of the U-shape, e.g., along an outlet leg of the U-shape extending from the housing outlet. In an example, the blower includes an axis that is generally co-linear with an axis of the outlet leg of the U-shape. In an example, the U-shaped air flow path extends substantially in the same plane.

(79) The RPT device **8000** is configured and structured to reduce noise output of the RPT device **8000** while maintaining a relatively small size.

(80) In the illustrated example, the RPT device **8000** provides two chambers, i.e., first chamber **8001** and second chamber **8002**. The first chamber **8001** is relatively large compared to the second chamber **8002**. As illustrated, the blower **8030** is supported in the first chamber **8001**, and receives air at the blower inlet **8032** from the second chamber **8002** (i.e., blower and blower inlet thereof located downstream of the chambers). The first and second plate assemblies **8050**, **8060** define at least a portion of the first and second chambers **8001**, **8002**, and each of the first and second plate assemblies **8050**, **8060** include a blower suspension **8054**, **8064** that supports the blower **8030** within the first chamber **8001** and separates and seals air flow through the first chamber **8001** from air flow through an interior of the blower **8030**. In addition, each of the first and second plate assemblies **8050**, **8060** include at least one tube (e.g., inlet tube array **8052** and flow tube array **8062**, respectively) such that air enters and exits the first chamber **8001** via at least one tube to reduce noise.

(81) In the illustrated example best shown in FIG. **4E**, the air flow path of the RPT device **8000** is structured and arranged such that air enters the housing **8010** via the housing inlet **8018**, passes through the inlet tube array **8052** provided by the first plate assembly **8050**, and into the first chamber **8001**. The first chamber **8001** receives air from the inlet tube array **8052** and delivers the

air to the flow tube array **8062** provided to the second plate assembly **8060**. The air passes through the flow tube array **8062** and into the second chamber **8002**. The second chamber **8002** receives air from the flow tube array **8062** and delivers the air to the blower inlet **8032** of the blower **8030**. The air flows through the blower **8030** such that a flow of air at positive pressure is provided at the blower outlet **8034** of the blower **8030**, which pressurized air exits the housing **8010** via the housing outlet **8020**.

(82) In an example, a flow rate sensor may be provided to the RPT device **8000** and structured and configured to measure a first pressure in the first chamber **8001** and a second pressure in the second chamber **8002** to determine an air flow rate. The first pressure and the second pressure may be used to determine a flow rate based on a configuration of the flow tube array **8062**, such as a pressure drop and/or the aerodynamic impedance.

(83) 5.4.1 RPT Device Mechanical & Pneumatic Components

(84) An RPT device may comprise one or more of the following components in an integral unit. In an alternative form, one or more of the following components may be located as respective separate units.

(85) 5.4.1.1 Housing

(86) As best shown in FIGS. **4H** to **4K**, the top housing portion **8012** of the housing **8010** includes a first part or base mold **8012A** constructed of a relatively rigid material (e.g., polypropylene) and a second part or overmold **8012B** constructed of a relatively soft material (e.g., thermoplastic elastomer (TPE) or silicone) that is provided (e.g., by overmolding) to the first part **8012A**. In the illustrated example, the overmold **8012B** is provided to an exterior surface of the base mold **8012A**, however, it should be appreciated that the overmold **8012B** may be provided to interior and/or exterior surfaces of the base mold **8012A**. Softness of the overmold **8012B** may provide a desirable tactile quality to a user of the RPT device **8000**. The overmold **8012B** may comprise a highly damped material to improve shock resistance and provide damping properties to attenuate wall-radiated noise.

(87) As best shown in FIGS. **4A**, **4B**, **4H**, **4J**, and **4K**, the top housing portion **8012** includes an on/off or power button **8003** and a wireless (e.g., Bluetooth) connection button **8004**, each of which is structured and configured to interact with the PCBA **8070**. However, it should be appreciated that the RPT device **8000** may include additional and/or alternative input devices to allow a user to interact with the device, e.g., one or more buttons, switches, dials, touch screen.

(88) In the illustrated example, as best shown in FIGS. **4H**, **4J**, and **4K**, the base mold **8012A** provides a cantilevered button portion **8003A** of the power button **8003** with a groove around the side of the button portion **8003A** that allows the button portion **8003A** to flex with respect to the base mold **8012A**. The overmold **8012B** provides a button portion **8003B** including a raised portion of the power button **8003** and webbing within the groove around the side of the button portion **8003A**. The raised portion and webbing of the button portion **8003B** provide a soft tactile feel for ease of use and grip and spring (button return) force. With respect to the wireless (Bluetooth) connection button **8004**, the base mold **8012A** provides an opening **8004A** and the overmold **8012B** provides a button portion **8004B** for a soft tactile feel for ease of use and grip and spring (button return) force.

(89) The bottom housing portion **8014** of the housing **8010** includes a first part or base mold **8014A** constructed of a relatively rigid material (e.g., polypropylene) and a second part or overmold **8014B** constructed of a relatively soft material (e.g., TPE or silicone) that is provided (e.g., by overmolding) to the first part **8014A**. In the illustrated example, the overmold **8014B** is provided to an exterior surface of the base mold **8014A**, however, it should be appreciated that the overmold **8014B** may be provided to interior and/or exterior surfaces of the base mold **8014A**. Softness of the overmold **8014B** may provide a desirable tactile quality to a user of the RPT device **8000**. The overmold **8014B** may comprise a highly damped material to improve shock resistance and provide damping properties to attenuate wall-radiated noise.

(90) As best shown in FIGS. 4F, 4G, and 4L, the bottom housing portion **8014** includes interior slots **8015A** (e.g., formed by spaced apart side walls) along one end thereof structured and configured to receive and support the first plate assembly **8050** and interior slots **8015B** (e.g., formed by spaced apart side walls) along an opposing end thereof structured and configured to receive and support the second plate assembly **8060**.

(91) Interior blower supports, such as interior ribs **8017**, constructed of a relatively soft, high damping material (e.g., TPE or silicone) are provided (e.g., by overmolding) to the bottom housing portion **8014**, e.g., see FIGS. 4F and 4I. The interior ribs **8017** are axially spaced between the slots **8015A**, **8015B** and are structured and configured to at least partially surround the blower **8030** to provide support, shock resistance and/or damping properties for the blower **8030**. One or more additional interior blower supports, such as additional ribs may be provided to the housing at alternative locations, e.g., to provide shock resistance and/or damping properties. It is contemplated that interior blower supports may be in alternative forms, for example as bumps or plates.

(92) The housing **8010** of the RPT device **8000** may comprise a guiding member to facilitate alignment and connection of the RPT device **8000** with one or more complementary components of a respiratory treatment system, such as a humidifier **5000** and/or an external battery **9000** for powering the RPT device **8000**. For example, the bottom housing portion **8014** includes a rail **8019** on each side thereof. In an example, the rails **8019** may also facilitate alignment and connection of the top and bottom housing portions **8012**, **8014** and/or facilitate handling/grip of the RPT device **8000**.

(93) Each rail **8019** may define or comprise a movement path for the RPT device **8000** in relation to the complementary component, such as in a form of an elongate rectangular indent as shown in FIG. 4H or an elongate, longitudinal protrusion as shown in FIG. 4X1. Other forms of course may be also suitable.

(94) One or more of the rails **8019** may comprise a retention mechanism (e.g., a recessed slot **8110** as shown in FIGS. 4X1 and 4X2) for latching the RPT device **8000**, which may also be used to disengage the RPT device from the complementary component connected thereto.

(95) In some forms, each rail **8019** may be configured for relative movement against a complementary roller, bearing or a guide. It will be understood that alternatively, an RPT device **8000** may comprise a roller, bearing, or a guide for engagement to a rail located on a complementary component.

(96) The top and bottom housing portions **8012**, **8014** may be connected to one another in any suitable manner, e.g., mechanical fasteners, mechanical interlock and/or snap-fit connection.

(97) The first end portion **8016** is supported by the top and bottom housing portions **8012**, **8014**, e.g., connected to the top and bottom housing portions **8012**, **8014** by mechanical fasteners, mechanical interlock and/or snap-fit connection. The first end portion **8016** is supported adjacent the first plate assembly **8050** such that the housing inlet **8018** is structured and configured to communicate with the inlet tube array **8052** of the first plate assembly **8050** and the housing outlet **8020** is structured and configured to communicate with an outlet bellows **8056** of the first plate assembly **8050**, as described in greater detail below.

(98) The first end portion **8016** includes a cover plate **8021** with a plurality of openings **8023** that provide the housing inlet **8018**. The plurality of openings **8023** allow sufficient airflow while preventing the ingress of larger objects. A recessed opening **8025** (e.g., see FIGS. 4F and 4G) is provided to the first end portion **8016** in communication with the plurality of openings **8023**. The recessed opening **8025** interfaces with the inlet tube array **8052** provided to the first plate assembly **8050**, such that air flows through the plurality of openings **8023** and the opening **8025** to the inlet tube array **8052** in communication with the first chamber **8001**.

(99) The housing outlet **8020** includes a tube portion **8026** structured and configured to receive and retain an end of the air circuit **4170**, e.g., cuff of an air delivery tube. The tube portion **8026** and the end of the air circuit may be connected to one another in any suitable manner, e.g., mechanical

interlock, snap-fit connection and/or friction-fit. The interior of the tube portion **8026** is structured and configured to communicate with the outlet bellows **8056** of the first plate assembly **8050**, which allows the end of the air circuit **4170** to engage and form a seal with the outlet bellows **8056** which is in communication with the blower outlet **8034** of the blower **8030**, thereby forming a seal for the air path. Further examples and details of the connection of the air circuit to an outlet of an RPT device are disclosed in U.S. Provisional Appln. No. 62/130,813, filed Mar. 10, 2015, which is incorporated herein by reference in its entirety.

(100) The first end portion **8016** also includes an opening or tube portion **8027** structured and configured to receive and retain an end of an electrical plug. The tube portion **8027** and the end of the electrical plug may be connected to one another in any suitable manner, e.g., mechanical interlock, snap-fit connection and/or friction-fit. The interior of the tube portion **8027** is structured and configured to communicate with an electrical socket **8080**, which allows the end of the electrical plug to electrically engage with the electrical socket **8080**, thereby allowing electrical power to be supplied to the RPT device **8000**.

(101) The second end portion **8022** is supported by the top and bottom housing portions **8012**, **8014**, e.g., connected to the top and bottom housing portions **8012**, **8014** by mechanical fasteners, mechanical interlock and/or snap-fit connection. The second end portion **8022** cooperates with an interior wall **8028** (i.e., provided by the bottom housing portion **8014** and the intermediate cover **8040**) to define an interior chamber portion **8006** (e.g., see FIG. 4I) structured and configured to receive the secondary PCB **8074**. The interior chamber portion **8006** is exterior the air path.

(102) 5.4.1.2 Blower

(103) In the illustrated example, the blower **8030** of the RPT device **8000** includes a three-stage design structured and configured for producing a flow, or a supply, of air at positive pressure up to 45-50 cmH.sub.2O, e.g., in the range of 2-50 cmH.sub.2O, e.g., 3-45 cmH.sub.2O, 4-30 cmH.sub.2O. However, in alternative examples, the blower **8030** may include a single stage design, a two stage design, or four or more stage designs.

(104) As best shown in FIG. 4I, the blower **8030** includes a housing **8031** including an axial air inlet (blower inlet) **8032** and axial air outlet (blower outlet) **8034** between which are located three stages with three corresponding impellers **8033A**, **8033B**, **8033C**, i.e., first and second impellers **8033A**, **8033B** positioned on one side of the motor **8035** and a third impeller **8033C** positioned on the other side of the motor **8035**. However, other suitable impeller arrangements are possible. Each impeller is followed by a set of stator vanes structured and configured to direct the air flow to the next stage.

(105) In the illustrated example, the blower **8030** is supported within the first chamber **8001** and the blower housing **8031** is relatively rigid and structured and configured to sealingly separate air flow through an interior of the blower **8030** from the first chamber **8001**. In an example, the housing **8031** may comprise a plurality of housing portions (e.g., first housing part including inlet **8032**, second housing part including outlet **8034**, and intermediate housing parts (e.g., stationary components providing stator vanes to direct air flow) that are connected to one another (e.g., welded) to form a substantially sealed structure.

(106) Further examples and details of the blower **8030** are described in PCT Patent Application Publication No. WO 2013/020167, which is incorporated herein by reference in its entirety.

(107) As described in greater detail below, the first plate assembly **8050** and the second plate assembly **8060** each include a blower suspension **8054**, **8064** that cooperate to support the blower **8030** within the housing **8010**, provide seals for the air path, isolate vibrations of the blower, and provide shock resistance. The blower suspension **8054**, **8064** may provide additional spring and damping to isolate vibrations and provide shock resistance. Specifically, the first plate assembly **8050** provides an outlet end suspension **8054** to support the blower **8030** adjacent the blower outlet **8034** and the second plate assembly **8060** provides an inlet end suspension **8064** to support the blower **8030** adjacent the blower inlet **8032**, i.e., a suspension is located at each end of the blower

8030.

(108) 5.4.1.3 Intermediate Cover

(109) The intermediate cover **8040** (also referred to as an intermediate housing portion of the housing) is supported between the top and bottom housing portions **8012**, **8014** of the housing **8010**. The intermediate cover **8040** is structured and configured to define at least a portion of the air path (e.g., define top and sides of the air path) and to define at least a portion of interior chamber portions exterior of the air path to receive the PCBA **8070**.

(110) Similar to the bottom housing portion **8014** described above, the intermediate cover **8040** includes interior slots **8042A** (e.g., formed by spaced apart side walls) along one end thereof structured and configured to receive and support the first plate assembly **8050** and interior slots **8042B** (e.g., formed by spaced apart side walls) along an opposing end thereof structured and configured to receive and support the second plate assembly **8060**, e.g., see FIGS. **4H** and **4L**. Accordingly, the intermediate cover **8040** cooperates with the bottom housing portion **8014** to support and retain the first and second plate assemblies **8050**, **8060** within the RPT device **8000**.

(111) In the illustrated example, the intermediate cover **8040** may constitute an upper portion and the bottom housing portion **8014** may constitute a lower portion, the upper and lower portions being engageable or separable in a generally normal direction with respect an axis of the inlet tubes **8055**/flow tubes **8065** of each of the first and second plate assemblies **8050**, **8060** and/or in a direction generally in plane or parallel with respect to the base plate **8051**, **8061**. In an example, the upper and lower portions are engageable or separable in a generally normal direction with respect an axis of at least one tube of the first plate assembly **8050**.

(112) The intermediate cover **8040** also includes an end wall **8028A** that cooperates with an end wall **8028B** of the bottom housing portion **8014** to define interior wall **8028**, e.g., see FIGS. **4I** and **4L**. Interior wall **8028** separates the air flow path from the interior chamber portion **8006** configured to receive the secondary PCB **8074**, e.g., see FIG. **4I**. Also, the main wall **8044** of the intermediate cover **8040** cooperates with the top housing portion **8012** to define an interior chamber portion **8007** to receive the main PCB **8072**, e.g., see FIG. **4I**. Accordingly, the intermediate cover **8040** defines interior chamber portions **8006**, **8007** exterior of the air path to receive the main and secondary PCBs **8072**, **8074**.

(113) The intermediate cover **8040**, along with the bottom housing portion **8014** and the first and second plate assemblies **8050**, **8060**, cooperate to define the top, bottom, and sides of the first and second chambers **8001**, **8002**, and therefore the pneumatic air path that extends to the blower inlet **8032** of the blower **8030**.

(114) In the illustrated example, the intermediate cover **8040** includes a first part or base mold **8040A** constructed of a relatively rigid material (e.g., polypropylene) and a second part or overmold **8040B** constructed of a relatively soft material (e.g., TPE or silicone) that is provided (e.g., by overmolding) to the first part **8040A**. In the illustrated example, the overmold **8040B** is provided to an interior surface of the base mold **8040A**, i.e., overmold **8040B** provided along the air flow path. The overmold **8040B** provides damping properties to attenuate wall-radiated noise. However, it should be appreciated that the overmold **8040B** may be provided to interior and/or exterior surfaces of the base mold **8040A**.

(115) Also, the overmold **8040B** extends along the edges of the intermediate cover **8040**, which cooperates with sealing (e.g., TPE or silicone) provided along side wall edges of the bottom housing portion **8014**, to seal the air path.

(116) 5.4.1.4 First Plate Assembly

(117) As best shown in FIGS. **4M**, **4N**, and **4I**, the first plate assembly **8050** includes a base plate **8051**, an inlet tube array **8052**, and a tube portion **8053** including a blower suspension (outlet end suspension) **8054** along one end and an outlet bellows **8056** along the opposite end. The tube portion **8053** includes an opening **8053A** (e.g., see FIG. **4N**) communicated with a pressure port **8058**. In addition, a sealing lip or sealing flange **8059** is provided along the edge or perimeter of the

base plate **8051**.

(118) In an example, the base plate **8051**, the inlet tube array **8052**, and the tube portion **8053** comprise a first part or base mold constructed of a relatively rigid material (e.g., polypropylene), and the blower suspension **8054**, outlet bellows **8056**, pressure port **8058**, and sealing lip **8059** comprise a second part or overmold constructed of a relatively soft material (e.g., TPE or silicone) that is provided (e.g., by overmolding) to the first part.

(119) As described above, the first plate assembly **8050** is supported between the intermediate cover **8040** and the bottom housing portion **8014**, i.e., base plate **8051** supported within interior slots **8042A**, **8015A** provided by the intermediate cover **8040** and the bottom housing portion **8014**. The base plate **8051** defines a wall of the first chamber **8001** and the sealing lip **8059** along the perimeter of the base plate **8051** provides a seal along the edge of the first chamber **8001**.

(120) The blower suspension **8054** of the first plate assembly **8050** is in the form of an outlet end suspension to support the blower **8030** adjacent the blower outlet **8034** of the blower **8030**. The outlet end suspension **8054** (e.g., constructed of an elastomeric material such as TPE or silicone) includes a first end portion **8054A** provided (e.g., overmolded) to the tube portion **8053**, a second end portion **8054B** engaged or otherwise secured to the blower outlet **8034** of the blower **8030**, and a radially outwardly extending gusset portion **8054C** between the first and second end portions **8054A**, **8054B**.

(121) The second end portion **8054B** may be secured to the blower **8030** in any suitable manner, e.g., wrap around an outlet flange provided to blower outlet **8034** as shown in FIG. 4I. The blower suspension **8054** seals the blower outlet **8034** to the tube portion **8053**, thereby sealing the air path for air exiting the blower outlet **8034** from the first chamber **8001**. Also, the gusset portion **8054C** of the blower suspension **8054** allows flexibility and relative movement to isolate vibrations of the blower **8030** and provide shock resistance.

(122) As described above, the outlet bellows **8056** of the first plate assembly **8050** is provided within the tube portion **8026** of the housing outlet **8020**, and is structured and configured to form a seal with an end of the air circuit **4170**, e.g., cuff of an air delivery tube. The outlet bellows **8056** (e.g., constructed of an elastomeric material such as TPE or silicone) includes an end portion **8056A** provided (e.g., overmolded) to the tube portion **8053** and a bellows portion **8056B** that curves radially inwardly from the end portion **8056A**. The bellows portion **8056B** is flexible to allow the end of the air circuit to engage and form a seal with the bellows portion **8056B**. As shown in FIG. 4M, the perimeter of the outlet bellows **8056B** may be provided with spaced part projections **8056C**, e.g., to add rigidity to the base of the outlet bellows **8056B**.

(123) The pressure port **8058**, which may be integral with the blower suspension **8054**, is structured and configured to interface or otherwise connect to a pressure sensor.

(124) The inlet tube array **8052** includes a plurality of inlet tubes **8055** structured and configured to extend from the base plate **8051** into the first chamber **8001**, e.g., inlet tubes **8055** extend generally perpendicular with respect to the base plate **8051**. In the illustrated example, the inlet tube array **8052** provides a first end portion **8052A** that slightly protrudes from one side of the base plate **8051** so as to interface with the recessed opening **8025** in the first end portion **8016** of the housing **8010**. As illustrated, the recessed opening **8025** includes shape that corresponds to a shape of the first end portion **8052A** along its outer perimeter. The second end portion **8052B** of the inlet tube array **8052** protrudes from the other side of the base plate **8051** so as to extend into the first chamber **8001**. Thus, the air flow path extends from the housing inlet **8018**, through the inlet tube array **8052**, and into the first chamber **8001**.

(125) This arrangement reduces noise output of the RPT device **8000** by increasing acoustic impedance through the inlet tubes **8055** while maintaining a high inertance. In an example, longer inlet tubes **8055** may be preferable for noise reduction (due to the higher inertance), and the length of the inlet tubes **8055** may be tuned or selected to match the specific noise frequency characteristics of the RPT device **8000**.

(126) In an example, the length of inlet tubes **8055** may be configured such that it does not adversely interact with any high-amplitude noise frequencies. For example, the length of inlet tubes **8055** may be tuned such that it does not coincide with $\frac{1}{4}$ or $\frac{1}{2}$ wavelengths of any peaks in the blower's noise spectrum. Peaks in the blower's noise spectrum (i.e., tonal peaks) may be caused by one or more of bearing defects, blade pass, turbulence, and structural resonance.

(127) In the illustrated example, each of the plurality of inlet tubes **8055** includes a non-circular cross-sectional shape (e.g., hexagonal cross-sectional shape), and the plurality of inlet tubes **8055** are arranged adjacent to one another to form the inlet tube array **8052**. That is, the inlet tubes **8055** are arranged adjacent to one another such that adjacent tubes share at least one side wall or side wall portion, i.e., adjacent tubes include at least one common side wall or side wall portion. As illustrated, adjacent tubes **8055** are separated by a thin wall to allow efficient packaging of the inlet tube array **8052**. The use of a plurality of relatively small inlet tubes **8055** may also further encourage the flow travelling therethrough to be laminar.

(128) In an example, each of the plurality of inlet tubes **8055** may comprise a draft angle to allow a mould tool to be extracted from the tubes during the moulding process. Such draft angle may be in either direction, i.e., converging away from the base plate **8051** or towards the base plate **8051**.

(129) In an example, each of the plurality of inlet tubes **8055** includes a length of about 35-55 mm (e.g., about 40-50 mm, e.g., about 43 mm), a wall thickness of about 1 mm, and a flat-to-flat distance (diameter) of about 3-5 mm (e.g., about 3.5-4.5 mm, e.g., about 4.33 mm). Thus the cross-sectional area of each hexagonal tube **8055** may be about 10-20 mm² (e.g., about 16.2 mm²), and the total cross-sectional area of all inlet tubes **8055** may be about 100-130 mm² (e.g., about 110-120 mm², e.g., about 114 mm²). However, it should be appreciated that other suitable lengths, wall thicknesses, flat-to-flat distances (diameters), and cross-sectional areas of the tubes **8055** are possible, e.g., depending on the desired noise characteristic.

(130) In the illustrated example, the inlet tube array **8052** includes seven inlet tubes **8055** arranged with six inlet tubes about a central inlet tube. However, it should be appreciated that other suitable number of tubes **8055** are possible (e.g., one or more inlet tubes, e.g., 5-10 inlet tubes) and the tubes **8055** may be arranged in other suitable manners (e.g., spaced apart, aligned in columns, etc.)

(131) Each of the plurality of inlet tubes **8055** may be constructed as a single part, or may comprise multiple parts. For example, each inlet tube **8055** may comprise an outer tube and an inlet restrictor configured to change a diameter of the tube.

(132) The plurality of inlet tubes **8055** may comprise inlet tubes of equal lengths and/or unequal lengths. The plurality of inlet tubes **8055** may comprise a sloped shape at one or more ends.

(133) Also, each of the plurality of inlet tubes **8055** may include other cross-sectional shapes, e.g., circular shape or noncircular shape (e.g., square, rectangle). In the illustrated example, the inlet tube array **8052** includes an outer perimeter with a non-circular shape. However, the outer perimeter shape of the inlet tube array **8052** may include other suitable shapes, e.g., circular shape or noncircular shape.

(134) For example, FIGS. **4S** to **4U** illustrate alternative arrangements for the inlet tube array **8052**. In FIG. **4S**, the inlet tube array **52A** includes a circular outer perimeter with each inlet tube including a square or truncated-square shape. In FIG. **4T**, the inlet tube array **52B** includes a circular outer perimeter with each inlet tube including a hexagon or truncated-hexagon shape. In FIG. **4U**, the inlet tube array **52C** includes a square outer perimeter with each inlet tube including a square shape.

(135) 5.4.1.5 Second Plate Assembly

(136) As best shown in FIGS. **4O** and **4P**, the second plate assembly **8060** includes a base plate **8061**, a flow tube array **8062**, and a blower suspension (outlet end suspension) **8064** supported within an opening **8063** provided to the base plate **8061**. In addition, a sealing lip or sealing flange **8069** is provided along the edge or perimeter of the base plate **8061**.

(137) In an example, the base plate **8061** and the flow tube array **8062** comprise a first part or base

mold constructed of a relatively rigid material (e.g., polypropylene), and the blower suspension **8064** and sealing lip **8069** comprise a second part or overmold constructed of a relatively soft material (e.g., TPE or silicone) that is provided (e.g., by overmolding) to the first part.

(138) As described above, the second plate assembly **8060** is supported between the intermediate cover **8040** and the bottom housing portion **8014**, i.e., base plate **8061** supported within interior slots **8042B**, **8015B** provided by the intermediate cover **8040** and the bottom housing portion **8014**. The base plate **8061** defines a wall of the first chamber **8001** and the second chamber **8002**, and the sealing lip **8069** along the perimeter of the base plate **8061** provides a seal along the edge of the first and second chambers **8001**, **8002**.

(139) The blower suspension **8064** of the second plate assembly **8060** is in the form of an inlet end suspension to support the blower **8030** adjacent the blower inlet **8032** of the blower **8030**. The inlet end suspension **8064** (e.g., constructed of an elastomeric material such as TPE or silicone) includes a radially outer portion **8064A** provided (e.g., overmolded) to the opening **8063** of the base plate **8061**, a radially inner portion **8064B** engaged or otherwise secured to the blower inlet **8032** of the blower **8030**, and an intermediate portion **8064C** between the outer and inner portions **8064A**, **8064B**.

(140) The radially inner portion **8064B** may be secured to the blower **8030** in any suitable manner, e.g., wrap around an inlet flange provided to the blower inlet **8032** as shown in FIG. 4I. The blower suspension **8064** provides a seal along the blower inlet **8032**, thereby sealing the blower inlet **8032** from the first chamber **8001** and providing an air path for air entering the blower inlet **8032** from the second chamber **8002**. Also, in the illustrated example, the intermediate portion **8064C** of the blower suspension **8064** is axially offset from the outer and inner portions **8064A**, **8064B**, which allows flexibility and relative movement to isolate vibrations of the blower **8030** and provide shock resistance.

(141) The flow tube array **8062** includes a plurality of flow tubes **8065** structured and configured to extend from the base plate **8061** into the first chamber **8001**, e.g., flow tubes **8065** extend generally perpendicular with respect to the base plate **8061**. In the illustrated example, each flow tube **8065** includes a first end portion **8065A** provided to the base plate **8061** and a second end portion **8065B** that protrudes from the base plate **8061** so as to extend into the first chamber **8001**. Thus, the air flow path extends from first chamber **8001**, through the flow tube array **8062**, and into the second chamber **8002**.

(142) Similar to the inlet tube array **8052**, the flow tube array **8062** is configured and arranged to reduce noise output of the RPT device **8000** by increasing acoustic impedance through the flow tubes **8065** while maintaining a high inertance.

(143) In the illustrated example, the flow tube array **8062** includes six, spaced-apart flow tubes **8065** generally arranged in three columns of two tubes with the columns generally offset from one another. However, it should be appreciated that other suitable number of tubes **8065** are possible (e.g., one or more flow tubes, e.g., 4-10 flow tubes) and the tubes **8065** may be arranged in other suitable manners (e.g., aligned in rows and/or columns, circular arrangement, adjacent tubes engaged with one another, etc.).

(144) In the illustrated example, each of the plurality of flow tubes **8065** includes a circular cross-sectional shape, however it should be appreciated that each of the tubes **8065** may include other cross-sectional shapes, e.g., circular shape or noncircular shape.

(145) In an example, each of the plurality of flow tubes **8065** may include any suitable length, diameter, wall thickness, and cross-sectional area, e.g., depending on the desired noise characteristic. In an example, the length of the flow tubes **8065** may be tuned or selected to match the specific noise frequency characteristics of the RPT device **8000**.

(146) In an example, each of the plurality of flow tubes **8065** may comprise a draft angle to allow a mould tool to be extracted from the tubes during the moulding process. Such draft angle may be in either direction, i.e., converging away from the base plate **8061** or towards the base plate **8061**.

(147) The plurality of flow tubes **8065** may comprise flow tubes of equal lengths and/or unequal lengths. The plurality of flow tubes **8065** may comprise a sloped shape at one or more ends.

(148) 5.4.1.6 Arrangement of Inlet Tube Array and Flow Tube Array

(149) In the illustrated example, the inlet tube array **8052** and the flow tube array **8062** are structured and arranged with respect to one another within the first chamber **8001** to reduce noise output of the RPT device **8000**.

(150) As best shown in FIGS. **4Q** and **4R**, the inlet tube array **8052** is axially spaced from the flow tube array **8062**, and the tubes **8055** of the inlet tube array **8052** include axes that are arranged substantially parallel to axes of the tubes **8065** of the flow tube array **8062**, however the tubes **8055** of the inlet tube array **8052** are not arranged co-axially with the tubes **8065** of the flow tube array **8062**. This arrangement reduces an amount of noise that is radiated through the flow tube array **8062** and then directly to the inlet tube array **8052**.

(151) The axially offset arrangement may be provided by off-setting a central axis of the inlet tube array **8052** (i.e., center axis defined by combination of all inlet tubes **8055**) from a central axis of the flow tube array **8062** (i.e., center axis defined by combination of all flow tubes **8065**) and/or the axially offset arrangement may be provided by arranging the inlet tube array **8052** and the flow tube array **8062** such that one or more of the individual tubes **8055**, **8065** are not co-axial or axially aligned (i.e., axially offset) with one another. For example, FIG. **4R** shows an arrangement wherein each of tubes **8055** of the inlet tube array **8052** includes an axis that is axially offset from axis of each of the tubes **8065** of the flow tube array **8062**.

(152) In an in-line configuration as illustrated, the tubes **8055** of the inlet tube array **8052** may be axially spaced from the tubes **8065** of the flow tube array **8062**, e.g., the spacing *d* (see FIG. **4Q**) between the inlet tube array **8052** and the flow tube array **8062** may be at least 5 mm, e.g., between about 10-15 mm.

(153) Also, in the illustrated example, the tubes **8055** of the inlet tube array **8052** and the tubes **8065** of the flow tube array **8062** each includes axes that are aligned substantially parallel with an axis of the blower **8030**.

(154) 5.4.1.7 Printed Circuit Board Assembly

(155) As described above, the PCBA **8070** includes a main PCB **8072** and a secondary PCB **8074**, the main PCB **8072** supported within interior chamber portion **8007** (e.g., defined by the top housing portion **8012** and the intermediate cover **8040**), and the secondary PCB **8074** supported within interior chamber portion **8006** (e.g., defined by the second end portion **8022**, the intermediate cover **8040**, and the bottom housing portion **8014**). This arrangement positions the main PCB **8072** and the secondary PCB **8074** exterior of the air flow path.

(156) The PCBA **8070** is electrically connected to the blower **8030** by one or more electrical connectors, e.g., electrical connector **8090** shown in FIGS. **4H** and **4L**. In the example shown in FIG. **4G**, the blower **8030** includes an electrical connector portion **8038** extending exterior the blower housing **8031**, and the electrical connector **8090** is structured and arranged to electrically connect the electrical connector portion **8038** of the blower **8030** to the PCBA **8070**.

(157) In an example, the electrical connector **8090** may be a flexible circuit board (FCB), flexible printed circuits (FPC) and/or flexible flat cables (FFC) to electrically connect the blower **8030** to the PCBA **8070**.

(158) In an example, the electrical connector **8090** may be arranged to pass through an internal air chamber (e.g., through the first chamber **8001**) of the RPT device **8000**, while not being positioned directly in the air flow path, e.g., such as between the inlet tube array **8052** and the flow tube array **8062**. For example, in the illustrated example, the electrical connector **8090** may be arranged to pass around a periphery of the inlet tube array **8052**.

(159) The electrical connector **8090** may be pre-formed to a particular shape to reduce potential interference with one or more components of the RPT device **8000**. Also, the electrical connector **8090** may be sufficiently long such that the electrical connector **8090** remains slack when

connected, e.g., to isolate vibration from being transmitted from the blower **8030** through the electrical connector **8090**.

(160) In an example, the electrical connector **8090** may be structured and configured to help achieve a reliable seal due to its low-profile form factor. In an example, tape and/or adhesives may be used to further seal any gaps where the electrical connector **8090** exits housing portions and/or intermediate cover.

(161) 5.4.1.8 Air Filter(s)

(162) An RPT device in accordance with one form of the present technology may include an air filter, or a plurality of air filters.

(163) In one form, an inlet air filter is located at the beginning of the pneumatic path upstream of the blower, such as housed in the cover plate **8021**.

(164) In one form, an outlet air filter, for example an antibacterial filter, is located between an outlet of the pneumatic block and a patient interface **3000**

(165) 5.4.1.9 Transducer(s)

(166) Transducers may be internal of the RPT device, or external of the RPT device. External transducers may be located for example on or form part of the air circuit, e.g., the patient interface. External transducers may be in the form of non-contact sensors such as a Doppler radar movement sensor that transmit or transfer data to the RPT device.

(167) In one form of the present technology, one or more transducers may be located upstream and/or downstream of the blower. The one or more transducers may be constructed and arranged to measure properties such as a flow rate, a pressure or a temperature at that point in the pneumatic path.

(168) In one form of the present technology, one or more transducers may be located proximate to the patient interface **3000**.

(169) In one form, a signal from a transducer may be filtered, such as by low-pass, high-pass or band-pass filtering.

(170) 5.4.1.9.1 Flow Rate Sensor

(171) A flow rate sensor in accordance with the present technology may be based on a differential pressure transducer, for example, an SDP600 Series differential pressure transducer from SENSIRION.

(172) In one form, a signal representing a flow rate such as a total flow rate Q_t from the flow rate sensor may be received by a central controller.

(173) 5.4.1.9.2 Pressure Sensor

(174) A pressure sensor in accordance with the present technology may be located in fluid communication with the pneumatic path. An example of a suitable pressure transducer is a sensor from the HONEYWELL ASDX series. An alternative suitable pressure transducer is a sensor from the NPA Series from GENERAL ELECTRIC.

(175) In one form, a signal from the pressure sensor may be received by the central controller.

(176) 5.4.1.9.3 Motor Speed Transducer

(177) In one form of the present technology a motor speed transducer may be used to determine a rotational velocity of the motor and/or the blower. A motor speed signal from the motor speed transducer may be provided to a therapy device controller. The motor speed transducer may, for example, be a speed sensor, such as a Hall effect sensor.

(178) 5.4.1.10 Anti-Spill Back Valve

(179) In one form of the present technology, an anti-spill back valve may be located between the humidifier **5000** and the pneumatic block. The anti-spill back valve is constructed and arranged to reduce the risk that water will flow upstream from the humidifier **5000**, for example to the motor of the blower.

(180) 5.4.1.11 Air Circuit

(181) An air circuit **4170** in accordance with an aspect of the present technology is a conduit or a

tube constructed and arranged in use to allow a flow of air to travel between two components such as the pneumatic block and the patient interface **3000**.

(182) In particular, the air circuit **4170** may be in fluid connection with the outlet of the pneumatic block and the patient interface. The air circuit may be referred to as an air delivery tube. In some cases there may be separate limbs of the circuit for inhalation and exhalation. In other cases a single limb is used.

(183) In some forms, the air circuit **4170** may comprise one or more heating elements configured to heat air in the air circuit, for example to maintain or raise the temperature of the air. The heating element may be in a form of a heated wire circuit, and may comprise one or more transducers, such as temperature sensors. In one form, the heated wire circuit may be helically wound around the axis of the air circuit **4170**. The heating element may be in communication with a controller such as a central controller. One example of an air circuit **4170** comprising a heated wire circuit is described in United States Patent Application No. US/2011/0023874, which is incorporated herewithin in its entirety by reference.

(184) 5.4.1.12 Oxygen Delivery

(185) In one form of the present technology, supplemental oxygen may be delivered to one or more points in the pneumatic path, such as upstream of the pneumatic block, to the air circuit **4170** and/or to the patient interface **3000**.

(186) 5.4.2 RPT Device Electrical Components

(187) 5.4.2.1 Power Supply

(188) A power supply may be located internal or external of the external housing of the RPT device **8000**.

(189) In one form of the present technology, power supply provides electrical power to the RPT device **8000** only. In another form of the present technology, power supply provides electrical power to both RPT device **8000** and humidifier **5000**.

(190) 5.4.2.2 Input Devices

(191) In one form of the present technology, an RPT device **8000** includes one or more input devices in the form of buttons, switches or dials to allow a person to interact with the device, e.g., buttons **8003**, **8004**. The buttons, switches or dials may be physical devices, or software devices accessible via a touch screen. The buttons, switches or dials may, in one form, be physically connected to the external housing, or may, in another form, be in wireless communication with a receiver that is in electrical connection to the central controller.

(192) In one form, the input device may be constructed and arranged to allow a person to select a value and/or a menu option.

(193) 5.4.2.3 Central Controller

(194) In one form of the present technology, the central controller is one or a plurality of processors suitable to control an RPT device **8000**, e.g., PCBA **8070** including main PCB **8072** and secondary PCB **8074**.

(195) Suitable processors may include an x86 INTEL processor, a processor based on ARM® Cortex®-M processor from ARM Holdings such as an STM32 series microcontroller from ST MICROELECTRONIC. In certain alternative forms of the present technology, a 32-bit RISC CPU, such as an STR9 series microcontroller from ST MICROELECTRONICS or a 16-bit RISC CPU such as a processor from the MSP430 family of microcontrollers, manufactured by TEXAS INSTRUMENTS may also be suitable.

(196) In one form of the present technology, the central controller is a dedicated electronic circuit.

(197) In one form, the central controller is an application-specific integrated circuit. In another form, the central controller comprises discrete electronic components.

(198) The central controller may be configured to receive input signal(s) from one or more transducers, one or more input devices, and the humidifier **5000**.

(199) The central controller may be configured to provide output signal(s) to one or more of an

output device, a therapy device controller, a data communication interface, and the humidifier
5000.

(200) In some forms of the present technology, the central controller is configured to implement the one or more methodologies described herein, such as the one or more algorithms expressed as computer programs stored in a non-transitory computer readable storage medium, such as memory. In some forms of the present technology, the central controller may be integrated with an RPT device. However, in some forms of the present technology, some methodologies may be performed by a remotely located device. For example, the remotely located device may determine control settings for a ventilator or detect respiratory related events by analysis of stored data such as from any of the sensors described herein.

(201) 5.4.2.4 Clock

(202) The RPT device may include a clock that is connected to the central controller.

(203) 5.4.2.5 Therapy Device Controller

(204) In one form of the present technology, the blower may be under the control of a therapy device controller. The therapy device controller may be a therapy control module that forms part of the algorithms executed by the central controller.

(205) In one form of the present technology, therapy device controller is a dedicated motor control integrated circuit. For example, in one form a MC33035 brushless DC motor controller, manufactured by ONSEMI is used.

(206) 5.4.2.6 Protection Circuits

(207) The one or more protection circuits in accordance with the present technology may comprise an electrical protection circuit, a temperature and/or pressure safety circuit.

(208) 5.4.2.7 Memory

(209) In accordance with one form of the present technology the RPT device includes memory, e.g., non-volatile memory. In some forms, memory may include battery powered static RAM. In some forms, memory may include volatile RAM.

(210) Memory may be located on the PCBA. Memory may be in the form of EEPROM, or NAND flash.

(211) Additionally or alternatively, RPT device includes a removable form of memory, for example a memory card made in accordance with the Secure Digital (SD) standard.

(212) In one form of the present technology, the memory acts as a non-transitory computer readable storage medium on which is stored computer program instructions expressing the one or more methodologies described herein, such as the one or more algorithms.

(213) 5.4.2.8 Data Communication Systems

(214) In one form of the present technology, a data communication interface is provided, and is connected to the central controller. Data communication interface may be connectable to a remote external communication network and/or a local external communication network. The remote external communication network may be connectable to a remote external device. The local external communication network may be connectable to a local external device.

(215) In one form, data communication interface is part of the central controller. In another form, data communication interface is separate from the central controller, and may comprise an integrated circuit or a processor.

(216) In one form, remote external communication network is the Internet. The data communication interface may use wired communication (e.g. via Ethernet, or optical fibre) or a wireless protocol (e.g. CDMA, GSM, LTE) to connect to the Internet.

(217) In one form, local external communication network utilises one or more communication standards, such as Bluetooth, or a consumer infrared protocol.

(218) In one form, remote external device is one or more computers, for example a cluster of networked computers. In one form, remote external device may be virtual computers, rather than physical computers. In either case, such a remote external device may be accessible to an

appropriately authorised person such as a clinician.

(219) The local external device may be a personal computer, mobile phone, tablet or remote control.

(220) 5.4.2.9 Output Devices Including Optional Display, Alarms

(221) An output device in accordance with the present technology may take the form of one or more of a visual, audio and haptic unit. A visual display may be a Liquid Crystal Display (LCD) or Light Emitting Diode (LED) display.

(222) 5.4.2.9.1 Display Driver

(223) A display driver receives as an input the characters, symbols, or images intended for display on the display, and converts them to commands that cause the display to display those characters, symbols, or images.

(224) 5.4.2.9.2 Display

(225) A display is configured to visually display characters, symbols, or images in response to commands received from the display driver. For example, the display may be an eight-segment display, in which case the display driver converts each character or symbol, such as the figure “0”, to eight logical signals indicating whether the eight respective segments are to be activated to display a particular character or symbol.

(226) 5.4.3 RPT Device Accessories

(227) 5.4.3.1 External Battery

(228) An external battery **9000** according to some forms of the present technology is shown in FIGS. **4W1** and **4W2**.

(229) The external battery **9000** may be engageable with the RPT device **8000**, for example as shown in FIGS. **4X1** to **4Y2**. The external battery **9000** may comprise a guiding member to facilitate alignment and connection with the RPT device **8000** (as well as disconnection). In one form, the external battery **9000** may comprise one or more guide slots **9050**, each configured to receive a respective guide rail **8019** of the RPT device **8000**. The guide slot **9050** may further comprise a latch **9060** releasable by a latch button **9065**. The latch **9060** may be configured to engage with the recessed slot **8110** along the guide rail **8019** of the RPT device **8000** to retain the RPT device **8000** with respect to the external battery **9000**.

(230) In some forms, the external battery **9000** may comprise a facia **9100** as shown in FIG. **4W1**, the facia being engageable with an end of the RPT device **8000** as shown in FIG. **4Y1**.

(231) The facia **9100** may comprise at least one of an air inlet **9018** and an air outlet **9020**, for extending an air path of the RPT device **8000**. The air inlet **9018** and/or the air outlet **9020** may be configured substantially identically to the housing inlet **8018** and the housing outlet **8020** respectively. Thus, a user may be able to use the same air conduit **4170** regardless of whether the air conduit is being connected to the RPT device **8000** or to the external battery **9000**.

(232) In some cases, the battery **9000** may comprise a muffler such that the noise output of the RPT device **8000** is reduced. For example, the muffler may reduce a noise output from the housing outlet **8020** so that it is reduced at the air outlet **9020**. Additionally, or alternatively, the muffler may reduce a noise output from the housing inlet **8018** so that noise from (e.g., measured at) the air inlet **9018** is lower than noise from (e.g., measured at) the housing inlet **8018**.

(233) In some examples, the battery facia **9100** may comprise a muffling chamber located between the housing outlet **8020** and the air outlet **9020**. In other examples, the battery facia **9100** may comprise a muffling chamber located between the housing inlet **8018** and the air inlet **9018**. To this end, the air outlet **9020** may be displaced from the housing outlet **8020**, e.g., by approximately 30 mm. It will be however understood that other means of noise reduction may be possible between the housing outlet **8020** and the air outlet **9020**, and the housing inlet **8018** and the air inlet **9018**.

(234) The external battery may comprise one or more battery cells (e.g., Lithium-Ion cells, or Nickel-metal hydride cells) configured to store electrical energy, and a printed circuit board assembly (PCBA) connected thereto.

(235) The PCBA may comprise components and/or circuitry for controlling one or more operations of the battery, such as power management, communication with the RPT device **8000** and/or voltage conversion.

(236) The external battery may generate heat, such as from the PCBA and/or the battery cells. Thus, it may be preferred to manage heat output from the external battery such that the PCBA and the battery cells operate within their preferred environmental conditions.

(237) In one aspect of the present technology, the external battery **9000** may be configured to be thermally coupled to the RPT device **8000**. The RPT device **8000** comprises an air path within, wherein ambient air is drawn in through its inlet (e.g., housing inlet **8018**) and delivered through its outlet (e.g., housing outlet **8020**).

(238) Thus, the air flow of the RPT device **8000** may be used to cool one or more components of the external battery **9000**. That is, heat conducted from the external battery **9000** to the RPT device **8000** may be removed from the RPT device **8000** by convection.

(239) The external battery **9000** may be configured such that its generated heat is more efficiently delivered to the RPT device **8000** for convective cooling. For example, higher heat generating components of the external battery **9000** may be located proximal to the RPT device **8000** when assembled. In another example, the external battery **9000** may comprise a thermally conductive element configured to deliver heat towards the RPT device **8000** when assembled.

(240) In one form, such as shown in FIG. **4Y1** wherein the external battery **9000** is configured to couple below the RPT device, a higher heat generating component of the external battery **9000** may be located towards the top of the external battery to improve heat transfer. In another form, the external battery **9000** may comprise a heat pipe to improve thermal conduction from a higher heat generating component to an exterior of the external battery **9000**, such that heat transfer to the RPT device **8000** may be improved when they are coupled together.

(241) Furthermore, the external battery **9000** may be configured to improve thermal conductivity between it and the RPT device **8000**. For example, the external battery **9000** may comprise one or more conductive portions (e.g., surfaces) configured to thermally couple with the RPT device **8000** for heat transfer when engaged thereto.

(242) 5.4.4 RPT Device Algorithms

(243) 5.4.4.1 Pre-Processing Module

(244) As shown in FIG. **4V**, a pre-processing module **4310** in accordance with one form of the present technology receives as an input a signal from a transducer, for example a flow rate sensor or pressure sensor, and performs one or more process steps to calculate one or more output values that will be used as an input to another module, for example a therapy engine module **4320**.

(245) In one form of the present technology, the output values include the interface or mask pressure P_m , the respiratory flow rate Q_r , and the leak flow rate Q_l .

(246) In various forms of the present technology, the pre-processing module **4310** comprises one or more of the following algorithms: pressure compensation **4312**, vent flow rate estimation **4314**, leak flow rate estimation **4316**, and respiratory flow rate estimation **4318**.

(247) 5.4.4.2 Therapy Control Module

(248) Therapy control module **4330** in accordance with one aspect of the present technology receives as inputs the therapy parameters from the therapy parameter determination algorithm of the therapy engine module **4320**, and controls the pressure generator to deliver a flow of air in accordance with the therapy parameters.

(249) In one form of the present technology, the therapy parameter is a treatment pressure P_t , and the therapy control module **4330** controls the pressure generator to deliver a flow of air whose mask pressure P_m at the patient interface **3000** is equal to the treatment pressure P_t .

5.5 Humidifier

(250) 5.5.1 Humidifier Overview

(251) In one form of the present technology there is provided a humidifier **5000** (e.g. as shown in

FIG. 5A) to change the absolute humidity of air or gas for delivery to a patient relative to ambient air. Typically, the humidifier **5000** is used to increase the absolute humidity and increase the temperature of the flow of air (relative to ambient air) before delivery to the patient's airways. (252) The humidifier **5000** may comprise a humidifier reservoir **5110**, a humidifier inlet **5002** to receive a flow of air, and a humidifier outlet **5004** to deliver a humidified flow of air. In some forms, as shown in FIG. 5A and FIG. 5B, an inlet and an outlet of the humidifier reservoir **5110** may be the humidifier inlet **5002** and the humidifier outlet **5004** respectively. The humidifier **5000** may further comprise a humidifier base **5006**, which may be adapted to receive the humidifier reservoir **5110** and comprise a heating element **5240**.

(253) 5.5.2 Humidifier Mechanical Components

(254) 5.5.2.1 Water Reservoir

(255) According to one arrangement, the humidifier **5000** may comprise a water reservoir **5110** configured to hold, or retain, a volume of liquid (e.g. water) to be evaporated for humidification of the flow of air. The water reservoir **5110** may be configured to hold a predetermined maximum volume of water in order to provide adequate humidification for at least the duration of a respiratory therapy session, such as one evening of sleep. Typically, the reservoir **5110** is configured to hold several hundred millilitres of water, e.g. 300 millilitres (ml), 325 ml, 350 ml or 400 ml. In other forms, the humidifier **5000** may be configured to receive a supply of water from an external water source such as a building's water supply system.

(256) According to one aspect, the water reservoir **5110** is configured to add humidity to a flow of air from the RPT device **4000** as the flow of air travels therethrough. In one form, the water reservoir **5110** may be configured to encourage the flow of air to travel in a tortuous path through the reservoir **5110** while in contact with the volume of water therein.

(257) According to one form, the reservoir **5110** may be removable from the humidifier **5000**, for example in a lateral direction as shown in FIG. 5A and FIG. 5B.

(258) The reservoir **5110** may also be configured to discourage egress of liquid therefrom, such as when the reservoir **5110** is displaced and/or rotated from its normal, working orientation, such as through any apertures and/or in between its sub-components. As the flow of air to be humidified by the humidifier **5000** is typically pressurised, the reservoir **5110** may also be configured to prevent losses in pneumatic pressure through leak and/or flow impedance.

(259) 5.5.2.2 Conductive Portion

(260) According to one arrangement, the reservoir **5110** comprises a conductive portion **5120** configured to allow efficient transfer of heat from the heating element **5240** to the volume of liquid in the reservoir **5110**. In one form, the conductive portion **5120** may be arranged as a plate, although other shapes may also be suitable. All or a part of the conductive portion **5120** may be made of a thermally conductive material such as aluminium (e.g. approximately 2 mm thick, such as 1 mm, 1.5 mm, 2.5 mm or 3 mm), another heat conducting metal or some plastics. In some cases, suitable heat conductivity may be achieved with less conductive materials of suitable geometry.

(261) 5.5.2.3 Humidifier Reservoir Dock

(262) In one form, the humidifier **5000** may comprise a humidifier reservoir dock **5130** (as shown in FIG. 5B) configured to receive the humidifier reservoir **5110**. In some arrangements, the humidifier reservoir dock **5130** may comprise a locking feature such as a locking lever **5135** configured to retain the reservoir **5110** in the humidifier reservoir dock **5130**.

(263) 5.5.2.4 Water Level Indicator

(264) The humidifier reservoir **5110** may comprise a water level indicator **5150** as shown in FIG. 5A-5B. In some forms, the water level indicator **5150** may provide one or more indications to a user such as the patient **1000** or a care giver regarding a quantity of the volume of water in the humidifier reservoir **5110**. The one or more indications provided by the water level indicator **5150** may include an indication of a maximum, predetermined volume of water, any portions thereof,

such as 25%, 50% or 75% or volumes such as 200 ml, 300 ml or 400 ml.

(265) 5.5.3 Humidifier Electrical & Thermal Components

(266) The humidifier **5000** may comprise a number of electrical and/or thermal components such as those listed below.

(267) 5.5.3.1 Humidifier Transducer(s)

(268) The humidifier **5000** may comprise one or more humidifier transducers (sensors) **5210** instead of, or in addition to, transducers described above. Humidifier transducers **5210** may include one or more of an air pressure sensor **5212**, an air flow rate transducer **5214**, a temperature sensor **5216**, or a humidity sensor **5218** as shown in FIG. 5C. A humidifier transducer **5210** may produce one or more output signals which may be communicated to a controller such as the central controller and/or the humidifier controller **5250**. In some forms, a humidifier transducer may be located externally to the humidifier **5000** (such as in the air circuit **4170**) while communicating the output signal to the controller.

(269) 5.5.3.1.1 Pressure Transducer

(270) One or more pressure transducers **5212** may be provided to the humidifier **5000** in addition to, or instead of, a pressure sensor provided in the RPT device.

(271) 5.5.3.1.2 Flow Rate Transducer

(272) One or more flow rate transducers **5214** may be provided to the humidifier **5000** in addition to, or instead of, a flow rate sensor provided in the RPT device.

(273) 5.5.3.1.3 Temperature Transducer

(274) The humidifier **5000** may comprise one or more temperature transducers **5216**. The one or more temperature transducers **5216** may be configured to measure one or more temperatures such as of the heating element **5240** and/or of the flow of air downstream of the humidifier outlet **5004**. In some forms, the humidifier **5000** may further comprise a temperature sensor **5216** to detect the temperature of the ambient air.

(275) 5.5.3.1.4 Humidity Transducer

(276) In one form, the humidifier **5000** may comprise one or more humidity sensors **5218** to detect a humidity of a gas, such as the ambient air. The humidity sensor **5218** may be placed towards the humidifier outlet **5004** in some forms to measure a humidity of the gas delivered from the humidifier **5000**. The humidity sensor may be an absolute humidity sensor or a relative humidity sensor.

(277) 5.5.3.2 Heating Element

(278) A heating element **5240** may be provided to the humidifier **5000** in some cases to provide a heat input to one or more of the volume of water in the humidifier reservoir **5110** and/or to the flow of air. The heating element **5240** may comprise a heat generating component such as an electrically resistive heating track. One suitable example of a heating element **5240** is a layered heating element such as one described in the PCT Patent Application Publication No. WO 2012/171072, which is incorporated herewith by reference in its entirety.

(279) In some forms, the heating element **5240** may be provided in the humidifier base **5006** where heat may be provided to the humidifier reservoir **5110** primarily by conduction as shown in FIG. 5B.

(280) 5.5.3.3 Humidifier Controller

(281) According to one arrangement of the present technology, a humidifier **5000** may comprise a humidifier controller **5250** as shown in FIG. 5C. In one form, the humidifier controller **5250** may be a part of the central controller **4230**. In another form, the humidifier controller **5250** may be a separate controller, which may be in communication with the central controller.

(282) In one form, the humidifier controller **5250** may receive as inputs measures of characteristics (such as temperature, humidity, pressure and/or flow rate), for example of the flow of air, the water in the reservoir **5110** and/or the humidifier **5000**. The humidifier controller **5250** may also be configured to execute or implement humidifier algorithms and/or deliver one or more output

signals.

(283) As shown in FIG. 5C, the humidifier controller **5250** may comprise one or more controllers, such as a central humidifier controller **5251**, a heated air circuit controller **5254** configured to control the temperature of a heated air circuit **4170** and/or a heating element controller **5252** configured to control the temperature of a heating element **5240**.

5.6 Breathing Waveforms

(284) FIG. 6 shows a model typical breath waveform of a person while sleeping. The horizontal axis is time, and the vertical axis is respiratory flow rate. While the parameter values may vary, a typical breath may have the following approximate values: tidal volume, V_t , 0.5 L, inhalation time, T_i , 1.6 s, peak inspiratory flow rate, Q_{peak} , 0.4 L/s, exhalation time, T_e , 2.4 s, peak expiratory flow rate, Q_{peak} , -0.5 L/s. The total duration of the breath, T_{tot} , is about 4 s. The person typically breathes at a rate of about 15 breaths per minute (BPM), with Ventilation, $Vent$, about 7.5 L/minute. A typical duty cycle, the ratio of T_i to T_{tot} is about 40%.

5.7 Glossary

(285) For the purposes of the present technology disclosure, in certain forms of the present technology, one or more of the following definitions may apply. In other forms of the present technology, alternative definitions may apply.

(286) 5.7.1 General

(287) Air: In certain forms of the present technology, air may be taken to mean atmospheric air, and in other forms of the present technology air may be taken to mean some other combination of breathable gases, e.g. atmospheric air enriched with oxygen.

(288) Ambient: In certain forms of the present technology, the term ambient will be taken to mean (i) external of the treatment system or patient, and (ii) immediately surrounding the treatment system or patient.

(289) For example, ambient humidity with respect to a humidifier may be the humidity of air immediately surrounding the humidifier, e.g. the humidity in the room where a patient is sleeping. Such ambient humidity may be different to the humidity outside the room where a patient is sleeping.

(290) In another example, ambient pressure may be the pressure immediately surrounding or external to the body.

(291) In certain forms, ambient (e.g., acoustic) noise may be considered to be the background noise level in the room where a patient is located, other than for example, noise generated by an RPT device or emanating from a mask or patient interface. Ambient noise may be generated by sources outside the room.

(292) Automatic Positive Airway Pressure (APAP) therapy: CPAP therapy in which the treatment pressure is automatically adjustable, e.g. from breath to breath, between minimum and maximum limits, depending on the presence or absence of indications of SDB events.

(293) Continuous Positive Airway Pressure (CPAP) therapy: Respiratory pressure therapy in which the treatment pressure is approximately constant through a respiratory cycle of a patient. In some forms, the pressure at the entrance to the airways will be slightly higher during exhalation, and slightly lower during inhalation. In some forms, the pressure will vary between different respiratory cycles of the patient, for example, being increased in response to detection of indications of partial upper airway obstruction, and decreased in the absence of indications of partial upper airway obstruction.

(294) Flow rate: The volume (or mass) of air delivered per unit time. Flow rate typically refers to an instantaneous quantity unless stated otherwise. In some cases, a reference to flow rate will be a reference to a scalar quantity, namely a quantity having magnitude only. In other cases, a reference to flow rate will be a reference to a vector quantity, namely a quantity having both magnitude and direction. Flow rate may be given the symbol Q . 'Flow rate' is sometimes shortened to simply 'flow'.

(295) Patient: A person, whether or not they are suffering from a respiratory disease.

(296) Pressure: Force per unit area. Pressure may be measured in a range of units, including cmH.sub.2O, g-f/cm.sup.2, hectopascal. 1 cmH.sub.2O is equal to 1 g-f/cm.sup.2 and is approximately 0.98 hectopascal. In this specification, unless otherwise stated, pressure is given in units of cmH.sub.2O.

(297) Respiratory Pressure Therapy (RPT): The application of a supply of air to an entrance to the airways at a treatment pressure that is typically positive with respect to atmosphere.

(298) Seal: May be a noun form ("a seal") which refers to a structure, or a verb form ("to seal") which refers to the effect. Two elements may be constructed and/or arranged to 'seal' or to effect 'sealing' therebetween without requiring a separate 'seal' element per se.

(299) Sound Power: The energy per unit time carried by a sound wave. The sound power is proportional to the square of sound pressure multiplied by the area of the wavefront. Sound power is usually given in decibels SWL, that is, decibels relative to a reference power, normally taken as 10-12 watt.

(300) Sound Pressure: The local deviation from ambient pressure at a given time instant as a result of a sound wave travelling through a medium. Sound pressure is usually given in decibels SPL, that is, decibels relative to a reference pressure, normally taken as 20×10^{-6} (Pa), considered the threshold of human hearing.

(301) 5.7.2 Terms for RPT Devices

(302) Leak: The word leak will be taken to be an unintended flow of air. In one example, leak may occur as the result of an incomplete seal between a mask and a patient's face. In another example leak may occur in a swivel elbow to the ambient.

(303) Noise, conducted (acoustic): Conducted noise in the present document refers to noise which is carried to the patient by the pneumatic path, such as the air circuit and the patient interface as well as the air therein. In one form, conducted noise may be quantified by measuring sound pressure levels at the end of an air circuit.

(304) Noise, radiated (acoustic): Radiated noise in the present document refers to noise which is carried to the patient by the ambient air. In one form, radiated noise may be quantified by measuring sound power/pressure levels of the object in question according to ISO 3744.

(305) Noise, vent (acoustic): Vent noise in the present document refers to noise which is generated by the flow of air through any vents such as vent holes in the patient interface.

(306) 5.7.3 Materials

(307) Silicone or Silicone Elastomer: A synthetic rubber. In this specification, a reference to silicone is a reference to liquid silicone rubber (LSR) or a compression moulded silicone rubber (CMSR). One form of commercially available LSR is SILASTIC (included in the range of products sold under this trademark), manufactured by Dow Corning. Another manufacturer of LSR is Wacker. Unless otherwise specified to the contrary, an exemplary form of LSR has a Shore A (or Type A) indentation hardness in the range of about 35 to about 45 as measured using ASTM D2240

(308) Polycarbonate: a typically transparent thermoplastic polymer of Bisphenol-A Carbonate.

5.8 Other Remarks

(309) A portion of the disclosure of this patent document contains material which is subject to copyright protection. The copyright owner has no objection to the facsimile reproduction by anyone of the patent document or the patent disclosure, as it appears in Patent Office patent files or records, but otherwise reserves all copyright rights whatsoever.

(310) Unless the context clearly dictates otherwise and where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit, between the upper and lower limit of that range, and any other stated or intervening value in that stated range is encompassed within the technology. The upper and lower limits of these intervening ranges, which may be independently included in the intervening ranges, are also encompassed within the technology, subject to any specifically excluded limit in the stated range. Where the stated range

includes one or both of the limits, ranges excluding either or both of those included limits are also included in the technology.

(311) Furthermore, where a value or values are stated herein as being implemented as part of the technology, it is understood that such values may be approximated, unless otherwise stated, and such values may be utilized to any suitable significant digit to the extent that a practical technical implementation may permit or require it.

(312) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this technology belongs. Although any methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present technology, a limited number of the exemplary methods and materials are described herein.

(313) When a particular material is identified as being used to construct a component, obvious alternative materials with similar properties may be used as a substitute. Furthermore, unless specified to the contrary, any and all components herein described are understood to be capable of being manufactured and, as such, may be manufactured together or separately.

(314) It must be noted that as used herein and in the appended claims, the singular forms “a”, “an”, and “the” include their plural equivalents, unless the context clearly dictates otherwise.

(315) All publications mentioned herein are incorporated herein by reference in their entirety to disclose and describe the methods and/or materials which are the subject of those publications. The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as an admission that the present technology is not entitled to antedate such publication by virtue of prior invention. Further, the dates of publication provided may be different from the actual publication dates, which may need to be independently confirmed.

(316) The terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced.

(317) The subject headings used in the detailed description are included only for the ease of reference of the reader and should not be used to limit the subject matter found throughout the disclosure or the claims. The subject headings should not be used in construing the scope of the claims or the claim limitations.

(318) Although the technology herein has been described with reference to particular examples, it is to be understood that these examples are merely illustrative of the principles and applications of the technology. In some instances, the terminology and symbols may imply specific details that are not required to practice the technology. For example, although the terms “first” and “second” may be used, unless otherwise specified, they are not intended to indicate any order but may be utilized to distinguish between distinct elements. Furthermore, although process steps in the methodologies may be described or illustrated in an order, such an ordering is not required. Those skilled in the art will recognize that such ordering may be modified and/or aspects thereof may be conducted concurrently or even synchronously.

(319) It is therefore to be understood that numerous modifications may be made to the illustrative examples and that other arrangements may be devised without departing from the spirit and scope of the technology.

5.9 Reference Signs List

(320) TABLE-US-00004 Number Feature Item inlet tube array 52A inlet tube array 52B inlet tube array 52C patient 1000 bed partner 1100 patient interface 3000 seal - forming structure 3100 plenum chamber 3200 stabilising structure 3300 vent 3400 connection port 3600 forehead support 3700 RPT device 4000 air circuit 4170 pre-processing module 4310 pressure compensation algorithm 4312 vent flow rate estimation algorithm 4314 leak flow rate estimation algorithm 4316

flow rate estimation algorithm 4318 therapy engine module 4320 therapy control module 4330 humidifier 5000 humidifier inlet 5002 humidifier outlet 5004 humidifier base 5006 reservoir 5110 conductive portion 5120 humidifier reservoir dock 5130 locking lever 5135 water level indicator 5150 humidifier transducer 5210 pressure transducer 5212 flow rate transducer 5214 temperature transducer 5216 humidity sensor 5218 heating element 5240 humidifier controller 5250 central humidifier controller 5251 heating element controller 5252 air circuit controller 5254 RPT device 8000 first chamber 8001 second chamber 8002 power button 8003 button portion 8003A button portion 8003B bluetooth connection button 8004 opening 8004A button portion 8004B interior chamber portion 8006 interior chamber portion 8007 housing 8010 top housing portion 8012 base mold 8012A overmold 8012B bottom housing portion 8014 base mold 8014A overmold 8014B slot 8015A slot 8015B first end portion 8016 interior rib 8017 housing inlet 8018 rail 8019 housing outlet 8020 cover plate 8021 second end portion 8022 opening 8023 faceplate 8024 opening 8025 tube portion 8026 tube portion 8027 interior wall 8028 end wall 8028A end wall 8028B blower 8030 housing 8031 inlet 8032 impeller 8033A impeller 8033B impeller 8033C outlet 8034 motor 8035 electrical connector portion 8038 intermediate cover 8040 base mold 8040A overmold 8040B slot 8042A slot 8042B main wall 8044 first plate assembly 8050 base plate 8051 inlet tube array 8052 first end portion 8052A second end portion 8052B tube portion 8053 opening 8053A blower suspension 8054 first end portion 8054A second end portion 8054B gusset portion 8054C inlet tube 8055 outlet bellow 8056 end portion 8056A bellows portion 8056B projections 8056C pressure port 8058 sealing lip 8059 second plate assembly 8060 base plate 8061 flow tube array 8062 opening 8063 blower suspension 8064 outer portion 8064A inner portion 8064B intermediate portion 8064C flow tube 8065 first end portion 8065A second end portion 8065B sealing lip 8069 PCBA 8070 main PCB 8072 secondary PCB 8074 electrical socket 8080 electrical connector 8090 recessed slot 8110 external battery 9000 air inlet 9018 air outlet 9020 guide slot 9050 latch 9060 latch button 9065 facia 9100

Claims

1. Apparatus for generating a supply of air at positive pressure for the amelioration or treatment of a respiratory disorder, comprising: a housing including a housing inlet and a housing outlet, wherein the housing is configured to form at least a portion of at least one chamber; a blower supported within the housing and configured to produce a flow of air at positive pressure; and a plate assembly supported within the housing, wherein the plate assembly includes a base plate configured to form a wall of the at least one chamber, at least one tube provided to the base plate and configured to allow air to enter the at least one chamber, and a blower suspension provided to the base plate via an opening in the base plate, and wherein the blower suspension is configured to support an end of the blower within the housing, wherein the blower suspension includes a gusset portion configured to allow flexibility and relative movement to isolate vibrations of the blower and provide shock resistance, wherein the plate assembly comprises an integral unit in which the blower suspension comprises an overmolded connection to the base plate, and wherein the plate assembly forms a separate and distinct unit from the housing.
2. Apparatus according to claim 1, wherein the plate assembly is a first plate assembly and the apparatus further comprises a second plate assembly, wherein the at least one tube of the first plate assembly is an inlet tube configured to allow air to enter the at least one chamber and the second plate assembly includes at least one flow tube configured to allow air to exit the at least one chamber.
3. Apparatus according to claim 2, wherein the second plate assembly includes a base plate configured to form a wall of the at least one chamber.
4. Apparatus according to claim 3, wherein the at least one chamber includes a first chamber and a second chamber, and wherein the base plate of the first plate assembly forms a wall of the first

chamber and the base plate of the second plate assembly forms a wall of the first and second chambers.

5. Apparatus according to claim 1, wherein the plate assembly is a first plate assembly and the apparatus further comprises a second plate assembly, wherein the blower suspension of the first plate assembly is an outlet end suspension configured to support the blower adjacent a blower outlet of the blower and the second plate assembly includes an inlet end suspension configured to support the blower adjacent a blower inlet of the blower.

6. Apparatus according to claim 5, wherein the second plate assembly includes a base plate configured to form a wall of the at least one chamber.

7. Apparatus according to claim 6, wherein the at least one chamber includes a first chamber and a second chamber, and wherein the base plate of the first plate assembly forms a wall of the first chamber and the base plate of the second plate assembly forms a wall of the first and second chambers.

8. Apparatus according to claim 1, wherein the blower includes an axis that is substantially parallel with an axis of the at least one tube.

9. Apparatus according to claim 1, wherein the base plate and the at least one tube comprise a first part constructed of a relatively rigid material and the blower suspension comprises a second part constructed of a relatively soft material provided to the first part.

10. Apparatus according to claim 9, wherein the second part comprises an elastomeric material that is overmolded to the first part.

11. Apparatus according to claim 1, wherein the plate assembly further comprises an outlet bellows provided to the base plate and in communication with a blower outlet of the blower, the outlet bellows configured to form a seal with an end of an air circuit.

12. Apparatus according to claim 11, wherein the outlet bellows comprises an overmolded connection to the base plate.

13. Apparatus according to claim 1, wherein the plate assembly further comprises a sealing lip along a perimeter of the base plate configured to provide a seal along an edge of the at least one chamber.

14. Apparatus according to claim 13, wherein the sealing lip comprises an overmolded connection to the base plate.

15. Apparatus according to claim 1, wherein the plate assembly further comprises a pressure port configured to connect to a pressure sensor.

16. Apparatus according to claim 15, wherein the pressure port comprises an overmolded connection to the base plate.

17. System for treating a respiratory disorder, comprising: a patient interface structured and configured to form a seal with a patient's face; the apparatus according to claim 1; and an air circuit to connect the patient interface and the apparatus.

18. Apparatus according to claim 1, wherein the housing includes an upper portion and a lower portion, and each of the upper portion and the lower portion includes an interior slot configured to receive and support the plate assembly.

19. Apparatus according to claim 1, wherein the blower is located downstream of the at least one chamber.

20. Apparatus according to claim 1, wherein the housing inlet and the housing outlet are provided to a common side of the apparatus.

21. Apparatus according to claim 1, wherein the blower suspension forms an opening including an axis that is substantially parallel with an axis of the at least one tube.

22. Apparatus according to claim 1, wherein the at least one tube is configured to extend perpendicular with respect to the base plate.
